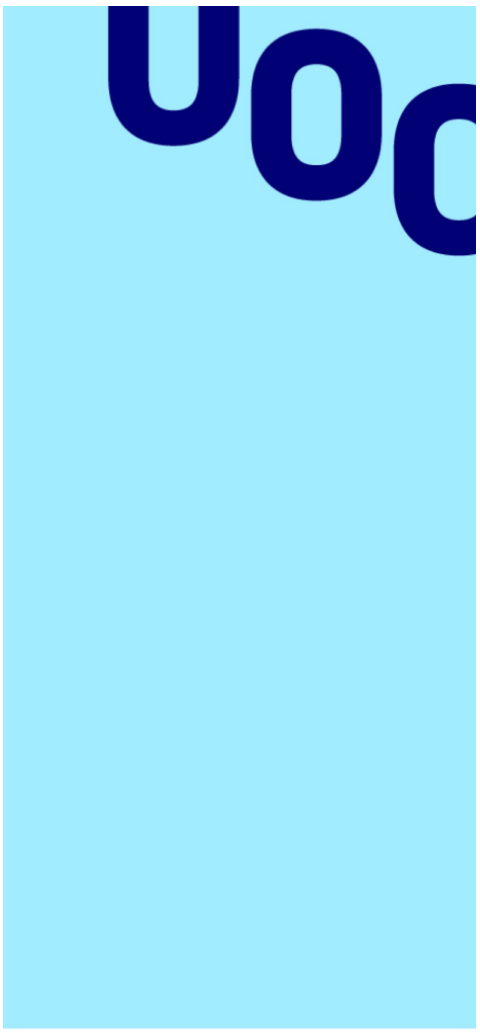
Comparative Evaluation of Satellite Image Segmentation Methods for Coastline Extraction: Spectral Indices vs. Deep Learning



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Area 4: Pattern Recognition and Computer Vision in Applied Sciences - Geographic Information Systems, satellite images and geopositioning

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Abstract

Accurate coastal monitoring is essential for managing erosion and assessing climate change impacts. While traditional satellite-derived shoreline extraction relies on spectral indices, these pixel-by-pixel methods often fail in highly dynamic macrotidal environments due to the semantic ambiguity caused by large wet sand areas in the intertidal zone. This Master's Thesis evaluates the operational viability of Deep Learning (DL) semantic segmentation as an alternative to traditional spectral thresholding for waterline extraction. Using Sentinel-2 imagery from two complex beaches in northern Spain (Salinas and El Puntal), a comprehensive benchmarking analysis was conducted. We compared state-of-the-art DL architectures (U-Net, DeepLabV3+, SegFormer) against widespread spectral indices (MNDWI, NDWI, AWEInsh, SCoWI). A robust Ground Truth dataset was generated to assess both pixel-level classification and sub-pixel geometric waterline accuracy. Results demonstrate that spectral indices suffer severe performance degradation and spatial biases due to inherent radiometric confusion. Conversely, DL models proved highly robust and geographically invariant, successfully interpreting the coastal context. The top-performing hybrid architecture (U-Net paired with an MiT-B2 transformer encoder) achieved a Mean Absolute Error of 7.27 m, halving the error of traditional methods and practically eliminating cross-shore biases. We conclude that AI-based workflows provide a superior, operational advantage for automated shoreline monitoring.

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Chapter 1

Introduction

1.1 Context and Justification of Work

Coastal regions are among the most dynamic, densely populated, and socio-economically vital areas on Earth. However, these environments are under constant threat from both natural processes and anthropogenic pressures, exacerbated by climate change. As sea levels rise and extreme weather events become more frequent, coastal erosion has emerged as a critical global challenge. Consequently, accurate and continuous monitoring of shoreline position (the instantaneous interface between land and water) is essential for understanding morphological changes, evaluating erosion trends, and planning effective coastal defense strategies(Vousdoukas et al., 2020).

Historically, tracking shoreline evolution relied on in-situ topographic surveys or aerial photogrammetry. While highly accurate, these methods are often prohibitively expensive, labor-intensive, and limited in spatial and temporal coverage. In recent decades, the advent of Earth Observation (EO) satellites has revolutionized coastal surveying. Missions such as the European Space Agency's (ESA) Sentinel-2 offer free, high-frequency, and multi-decadal imagery spanning the entire globe (Vitousek et al., 2023). This wealth of optical satellite data offers researchers and coastal managers to monitor shoreline dynamics at unprecedented scales and at a fraction of the cost.

To extract the exact position of the waterline from a satellite image, traditional methodologies rely heavily on spectral indices. These mathematical formulas exploit the differences in how water and land reflect specific bands of sunlight (such as the near-infrared) to create binary masks separating water from non-water. Frameworks utilizing these principles, such as CoastSat (Vos et al., 2019b), SHOREX (Sánchez-García et al., 2020), and CASSIE (Almeida et al., 2021), have become the operational standard for generating automated, large-scale shoreline extraction data worldwide, providing an efficient and widely-used alternative to manual digitization.

However, this traditional pixel-by-pixel approach faces severe limitations when applied to highly dynamic, macrotidal environments. In high-energy coasts with large tidal ranges (such as the Cantabrian Sea) the presence of extensive wet sand and breaking wave foam creates a persistent

"semantic ambiguity". The basic spectral premise that water strongly absorbs light while land reflects it frequently breaks down in these transition zones, leading to severe misidentifications of the actual waterline (see Figure 1.1). As demonstrated by recent studies (e.g., Cabezas-Rabadán et al., 2026), traditional tools struggle to navigate these challenging coastal scenes, resulting in significant spatial errors.

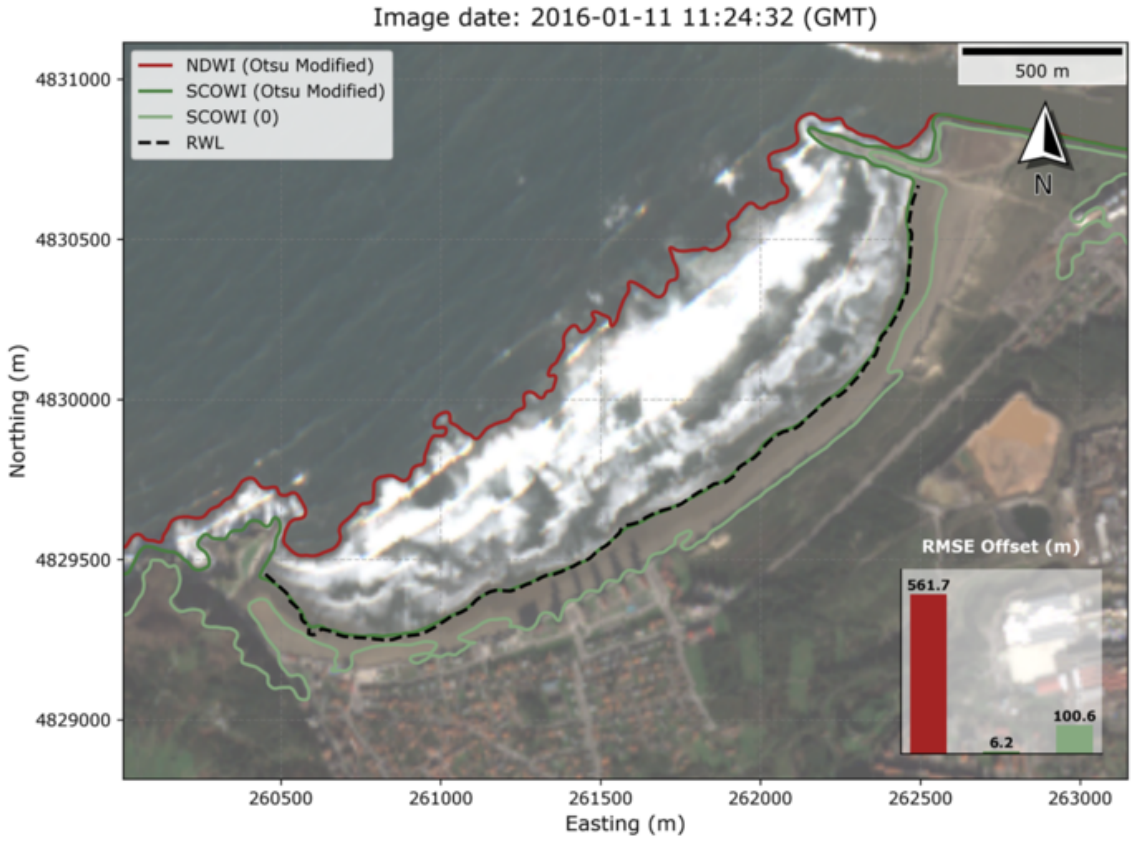


Figure 1.1: Examples of shoreline misidentification in a macrotidal beach (UTM ETRS89 30N) (adapted from Cabezas-Rabadán et al., 2026)

Consequently, there is a clear and urgent need to move beyond purely spectral formulas and explore methodologies capable of understanding the context of the image. Deep Learning (DL), a subset of Artificial Intelligence (AI), may offer a promising solution to bridge this gap. Unlike traditional index thresholds that evaluate the reflectance value of an isolated pixel, advanced DL semantic segmentation architectures can learn complex spatial patterns, textures, and contextual hierarchies. By emulating how the human eye interprets a satellite image (recognizing the texture of a wave or the gradient of wet sand) DL models hold the potential to overcome the spectral confusion inherent in macrotidal environments.

1.2 Work Objectives

Given the challenges posed by macrotidal coasts and the limitations of traditional spectral methods, the overarching goal of this Master's Thesis was to evaluate the operational viability of Deep Learning solutions for shoreline mapping. To ensure a robust assessment, the study focused on two highly dynamic beaches located along the northern Spanish coast: Salinas (Asturias) and El Puntal (Cantabria).

Primary Objective:

- To perform a comprehensive benchmarking analysis between traditional index-based thresholding (e.g., NDWI, MNDWI) and modern Deep Learning-based semantic segmentation for automated coastline extraction in complex coastal environments.

Secondary Objectives:

- To generate a high-quality "Ground Truth" dataset of coastal masks (classifying pixels into *water*, *white-water*, *wet sand*, and *sand*). This dataset was used exclusively to train and validate the Deep Learning segmentation models, and was rigidly constructed through manual photo-interpretation, utilizing AI-assisted annotation tools (such as the Segment Anything Model, SAM (Kirillov et al., 2023)) solely to refine and accelerate the process.
- To generate a high-quality "Ground Truth" dataset of photo-interpreted waterlines directly derived from the Sentinel-2 imagery. This dataset served primarily to benchmark the accuracy of the extracted waterlines (testing both traditional spectral indices-based models and Deep Learning segmentation models).
- To systematically evaluate and compare the performance degradation of both classical indices and Neural Networks under specifically adverse oceanographic and atmospheric conditions (e.g., the presence of extensive white foam and variations in intertidal sand moisture).

1.3 Impact on Sustainability, Ethics-Social and Diversity

In compliance with the UOC's cross-disciplinary competence "Ethical and global commitment", this section identifies the positive and/or negative impacts of the Master's Thesis on the three dimensions: Sustainability, Ethical Behavior and Social Responsibility, and Diversity and Human Rights.

1. Sustainability
The project used freely available Sentinel-2 Level-2A imagery from the Copernicus program and performed all processing locally. Traditional field-based coastal monitoring requires repeated physical surveys with vehicles, boats or aircraft, generating higher operational GHG emissions. The Deep Learning training phase incurred moderate energy consumption inherent to GPU usage; however, this was limited in scope and offset by the elimination of on-site campaigns. No other significant negative environmental impacts (resource depletion, waste generation or large-scale land use) were identified.

Overall impact: neutral to slightly positive.

2. Ethical Behavior and Social Responsibility (RS)
All data used were open-access satellite products with no personal, sensitive or identifiable information. Ground-truth masks were generated through manual photo-interpretation assisted only by the Segment Anything Model for refinement. Model performance was evaluated with transparent metrics (IoU, Dice, geometric error) and limitations under adverse conditions (foam, wet sand, clouds) were explicitly quantified and reported. The project did not produce automated decisions affecting individuals or communities. Potential risk of misinterpretation of shoreline outputs by end-users was mitigated by providing full error analysis.

Overall impact: neutral.

3. Diversity and Human Rights
The thesis was strictly technical and involved no human subjects, personal data, biometric information or gender-related variables. Development was carried out individually with no team composition. The resulting code, models and documentation used objective technical language and did not contain implicit biases. No positive or negative impacts on human rights, gender diversity or vulnerable groups were identified.

Overall impact: neutral.

1.4 Focus and Method Followed

To address the goal of automating and improving shoreline extraction from Sentinel-2 satellite imagery, several strategies were initially considered. One theoretical approach could involve developing a completely novel Deep Learning architecture from scratch, comprehensively designed for water-sand segmentation. Conversely, an entirely applied approach could rely on utilizing existing proprietary software or closed-source commercial solutions.

However, the chosen strategy for this thesis was an applied comparative benchmarking approach. Rather than inventing a new architecture, this work focused on adapting and fine-tuning existing, state-of-the-art open-source Deep Learning architectures (such as the U-Net with varying feature-extraction encoders) and rigorously evaluating them against widely adopted traditional baselines (e.g., spectral indices combined with Marching Squares sub-pixel contouring).

This comparative framework was selected to maximize both practical impact and scientific rigor. At present, the primary challenge in automated shoreline monitoring lies not in developing novel algorithms, but in validating existing state-of-the-art models through standardized coastal metrics. By embedding established semantic segmentation networks into a highly specialized evaluation pipeline, this work shifted the focus from model creation to operational viability and sub-pixel accuracy. Consequently, this ensured a transparent methodology capable of answering the fundamental research question (the operational superiority of Deep Learning over traditional spectral methods) without redundant efforts in foundational architectural design.

1.5 Work Schedule

The project was developed over a 5-month period (including the thesis defence), structured into the following tasks as shown in the Gantt chart (Figure 1.2). Additionally, project management software such as Trello or MS Planner were used to track and control the tasks at a more specific level:

- **Task 1:** Literature review, selection of study sites (beaches), and satellite images download.
  - Literature review on traditional methods for satellite-derived coastline extraction.
  - Literature review on Deep Learning models applied to coastal monitoring.
  - Satellite images download for northern Spanish coast beaches: Salinas (Asturias) and El Puntal (Cantabria).
- **Task 2:** Calculation of spectral indices and generation of Ground Truth datasets.
  - Preparation of the GitHub repository structure for the project's codebase.
  - Calculation of spectral indices (e.g., NDWI, MNDWI).
  - Create RGB composites to assist in the annotation process.
  - Ground Truth generation through manually photo-interpretation of the RGB composites.
  - Generation of Ground Truth coastline vectors directly from the Sentinel-2 images, to be used for benchmarking.
- **Task 3:** Model implementation: Development of baseline and Deep Learning architectures.
  - Application of automatic thresholding to spectral indices to set baseline models.
  - Selection, development, and training of Deep Learning architectures.
  - Application of contouring algorithms to extract shorelines at sub-pixel precision.
- **Task 4:** Statistical analysis of results, benchmarking across different algorithms, and final evaluation.
  - Pixel-based metrics evaluation (e.g., IoU, Dice Coefficient).

  - Geometric validation by calculating the error distance between predicted and real shorelines.
  - Comparison of performance under adverse conditions.

- **Task 5:** Drafting report and interim deliveries.
- **Task 6:** Progress meetings with directors.

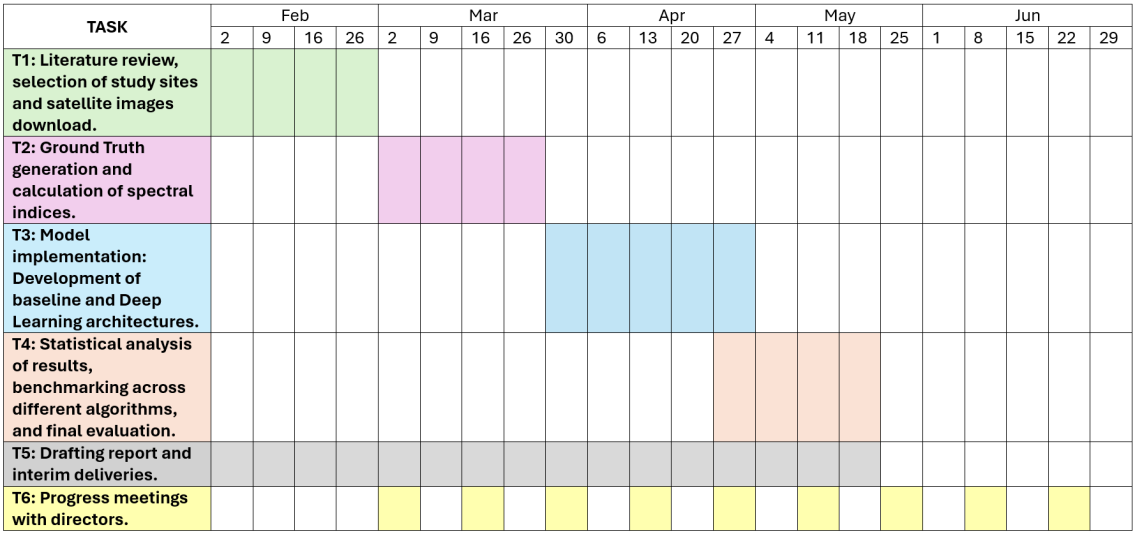


Figure 1.2: Gantt chart of the project timeline.

1.6 Brief Summary of Products Obtained

This project has yielded several tangible and reusable products:

- A comprehensive annotated dataset of macrotidal coastal images, encompassing 161 multi-class pixel-wise masks (categorizing water, white-water, wet sand, and dry sand) and 285 manually digitized reference vector waterlines derived directly from Sentinel-2 imagery.
- An open-source Python codebase hosted on a public GitHub repository, which includes the complete end-to-end processing pipeline: from satellite data preprocessing and spectral index calculation, to Deep Learning model training and advanced geometric benchmarking.
- A suite of fine-tuned and strictly evaluated state-of-the-art Deep Learning models (U-Net, DeepLabV3+, SegFormer) optimized specifically for challenging coastal dynamics.

1.7 Brief Description of Other Memory Chapters

The document is organized as follows:

- **Chapter 1. Introduction:** Provides the context and justification for the work, clearly defines the research objectives, outlines the expected impacts on sustainability, ethics, and diversity, describes the methodological approach, and presents a detailed project timeline.
- **Chapter 2. State of Art:** Reviews the evolution of coastal monitoring, the principles of satellite-based Earth Observation, the traditional use of spectral indices and their limitations, and the recent paradigm shift toward Deep Learning in remote sensing.
- **Chapter 3. Materials and Methods:** Details the geographical context of the study areas, the generation of the reference Ground Truth datasets, the implementation of both traditional and neural network pipelines, and the rigorous pixel-based and geometric evaluation frameworks.
- **Chapter 4. Results:** Presents the findings of the benchmarking process, analyzing the physical characteristics of the beaches, the internal semantic performance of the neural networks, and the overarching cross-shore geometric extraction errors.
- **Chapter 5. Conclusions and Future Works:** Synthesizes the core findings, explicitly answers the research objectives, details the foreseen impacts (sustainability, ethics, and diversity), and outlines actionable recommendations for future research.
- **Glossary:** Provides definitions for technical terms and acronyms used throughout the document.

1.8 Statement on the Use of Generative Artificial Intelligence

During the development of this Master's Thesis, Generative Artificial Intelligence tools (such as Gemini and GitHub Copilot) were utilized purely as assistive instruments. Their application was strictly limited to code debugging, algorithmic structuring, grammar correction, and structural prose refinement to improve the readability of the manuscript. The design of the scientific methodology, the manual photo-interpretation of the Ground Truth datasets, the critical evaluation of the results, and the core intellectual contributions remain exclusively the original work of the author.

Chapter 2

State of Art

2.1 Coastal Monitoring and Shoreline Definition

The analysis of shoreline evolution over time is a fundamental aspect of coastal science, engineering, and management. An idealized definition of the shoreline describes it as the physical interface between land and water (Boak and Turner, 2005). However, this boundary is highly dynamic, driven by continuous fluctuations in water levels caused by tides, waves, storm surges, and seasonal changes. Consequently, capturing a single, definitive shoreline position is practically impossible, making it necessary to define the shoreline based on specific objectives and available data.

To overcome this, researchers typically rely on *shoreline indicators*, which serve as proxy features to represent the true shoreline position. According to the comprehensive review by Boak and Turner (2005), these indicators are generally grouped into two main categories:

- **Visually discernible features**: Physical marks left on the coast, such as the High-Water Line (HWL) (often identified by a change in sand color due to moisture content), the instantaneous wet/dry boundary left by wave runup, or the seaward edge of dune vegetation (Figure 2.1).
- **Tidal datum-based indicators**: These define the shoreline as the intersection of a specific coastal profile with a predetermined vertical elevation, such as Mean High Water (MHW) or Mean Sea Level (MSL). While datum-based indicators provide a mathematically objective boundary, they often require high-resolution topographic data (e.g., DGPS or LiDAR) which can be hard to acquire for long-term or large-scale studies (Figure 2.1).

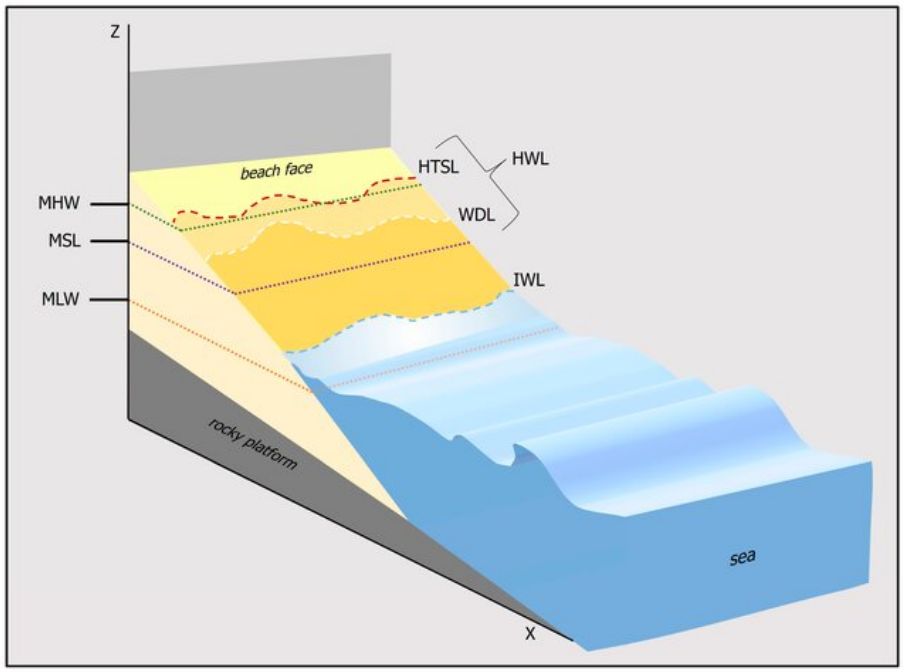


Figure 2.1: Schematic representation of a beach profile illustrating shoreline indicators.(1) Feature based, detected on beach face: HTSL—high-tide swash line; WDL—wet–dry line; HWL—high-water line; IWL—instantaneous water line. (2) Elevation-based: MHW—mean high-water, MSL—mean sea level; MLW—mean low-water. (Source: Valverde et al., 2025)

The selection of a specific indicator and monitoring frequency depends on the intended application of the data. Shoreline monitoring serves several critical functions:

- *Long-term Trend Analysis*: Quantifying multi-decadal erosion or accretion rates to understand coastline evolution.
- *Storm Impact Assessment*: Evaluating episodic erosion caused by extreme weather events and tracking the subsequent post-storm recovery.
- *Seasonal Variability Studies*: Capturing the cyclical movement of the shoreline due to changing wave climates between summer and winter.
- *Coastal Management*: Quantifying the dry beach width for coastal management purposes, such as evaluating the beach's natural potential to act as a coastal defense against storms, assessing its carrying capacity for tourism and recreational activities, and informing strategies to mitigate overall vulnerability.

Within this theoretical framework, it is essential to define the specific indicator addressed in this research. This Master's Thesis focuses exclusively on the instantaneous waterline (WL), defined as the precise boundary between the water body and the emerged land at the exact moment of satellite overpass. While the term *shoreline* is often used in a broader sense in coastal literature, throughout this document, both *shoreline* and *waterline* (or the acronym SDW for

Satellite-Derived Waterline) are used interchangeably to refer specifically to this instantaneous land-water interface. Focusing on the WL is particularly relevant for satellite-based monitoring, as it represents the most direct feature discernible from optical imagery, although its detection in macrotidal environments remains a significant technical challenge, as will be described in more detail in Section 2.5.

2.2 Traditional Coastal Monitoring Methods

Historically, coastal monitoring has relied heavily on in-situ physical surveys. While these methods can offer high precision, they are inherently constrained by significant logistical effort, high operational costs, and consequently, low spatial and temporal coverage.

Topobathymetric monitoring using Global Navigation Satellite Systems (GNSS), particularly those operating in Real-Time Kinematic (RTK) mode, have been widely employed to acquire cross-shore beach profiles (Morton et al., 1993). Although GNSS-based techniques can achieve millimeter-level accuracy, data collection is limited to the physical path traversed by operators (e.g., walking, driving, or mounted on a vehicle), making continuous shoreline mapping over regional scales largely impractical.

To bridge the spatial gap of ground surveys, researchers increasingly turned to advanced airborne scanning technologies such as Light Detection and Ranging (LiDAR). Airborne LiDAR allows for the rapid acquisition of dense, high-resolution 3D point clouds over hundreds of kilometers of coast, from which datum-based shorelines can be extracted objectively (Stockdon et al., 2002). However, the high economic cost of instrument deployment restricts its use to infrequent, opportunistic, or project-specific flights, failing to provide the high-frequency temporal data necessary for continuous monitoring.

Addressing the temporal limitations of physical surveys, fixed coastal video monitoring systems (most notably the Argus system (Holman et al., 1993)) were developed in the 1990s. These ground-based camera networks capture high-frequency images (e.g., every 10 minutes) and can produce time-averaged images to derive robust shoreline-proxy indicators. While providing excellent temporal resolution to track short-term morphological changes, their spatial coverage remains inextricably confined to the camera's visual field of view (typically 1–3 kilometers), leaving vast stretches of the coastline unmonitored.

More recently, the rise of ubiquitous smartphone technology has catalyzed the development of Citizen Science approaches for coastal monitoring, best exemplified by the global CoastSnap project (Harley and Kinsela, 2022). By installing simple, low-cost smartphone cradles at strategic coastal viewpoints, these initiatives crowdsource beach imagery from the general public. While this drastically reduces both initial deployment and operational costs while providing data with a spatial resolution similar to video stations, the temporal frequency and reliability of the data are inherently opportunistic, depending entirely on community participation and favorable weather conditions.

To overcome the inherent trade-offs between spatial coverage, temporal frequency, and operational costs present in these traditional methods, researchers have increasingly turned to

Earth Observation (EO) satellites. As summarized in Table 2.1, satellite imagery uniquely provides continuous, global-scale coverage with a high frequency of observations at virtually zero operational cost for the end-user, effectively resolving many of the historical monitoring limitations. The application of this remote sensing data for continuous shoreline extraction, which is the cornerstone of this thesis, will be discussed in detail in the following section (Section 2.3).

Table 2.1: Summary of common coastal monitoring methods and their key characteristics.

Monitoring Method	Spatial Scale	Spatial Resolution	Temporal Resolution	Operational Cost
GNSS / RTK Topography	Local	Millimetric	Low (Infrequent)	Very High (per survey)
Airborne LiDAR	Regional	Sub-meter	Low (Infrequent)	Very High (per flight)
Video systems (Argus)	Local (1-3 km)	Centimetric	Very High (Minutes)	Low (after initial setup)
Citizen Science (CoastSnap)	Local	Centimetric	Highly Variable	Very Low
Optical satellites (i.e. Landsat 4-9 and Sentinel-2)	Global	Decametric (10 m)	High (Days)	Zero (Free open data)

2.3 Satellite Remote Sensing for Coastal Monitoring

The advent of Earth Observation (EO) satellites has profoundly democratized and revolutionized coastal monitoring, effectively resolving the historical trade-offs between spatial coverage and temporal frequency. The cornerstone of this revolution is the Landsat program, jointly managed by NASA and the USGS. Initiated in the 1970s and providing continuous 30-meter resolution optical imagery since the launch of Landsat 4 in 1982, it offers an unparalleled, retrospective archive spanning over four decades. This extensive catalog allows coastal scientists and managers to reconstruct long-term shoreline evolution, analyze multi-decadal erosion trends, and assess the historical impact of major climatic events on a truly global scale.

Building upon this legacy, the European Space Agency (ESA) catalyzed a new era in EO with the launch of the Copernicus Sentinel-2 constellation. Initially a twin-satellite system (Sentinel-2A and 2B, launched in 2015 and 2017), the constellation has recently been reinforced with the launch of Sentinel-2C in September 2024. Sentinel-2 dramatically bridged the gap to traditional in-situ methods by providing freely available, multispectral imagery at an unprecedented 10-meter spatial resolution (in visible and Near-Infrared bands) with a high-frequency revisit time of just 5 days at the equator (Vitousek et al., 2023). This exponential leap in temporal resolution has unlocked the ability to monitor highly dynamic, short-term coastal processes from space, such as seasonal beach profile fluctuations (e.g., Warrick et al., 2025) and the acute morphodynamic impacts of specific storms (e.g., Cabezas-Rabadán et al., 2024). The synergistic combination of both Landsat and Sentinel-2 missions has sparked an exponential growth in satellite-derived shoreline research over the last decade (Figure 2.2).

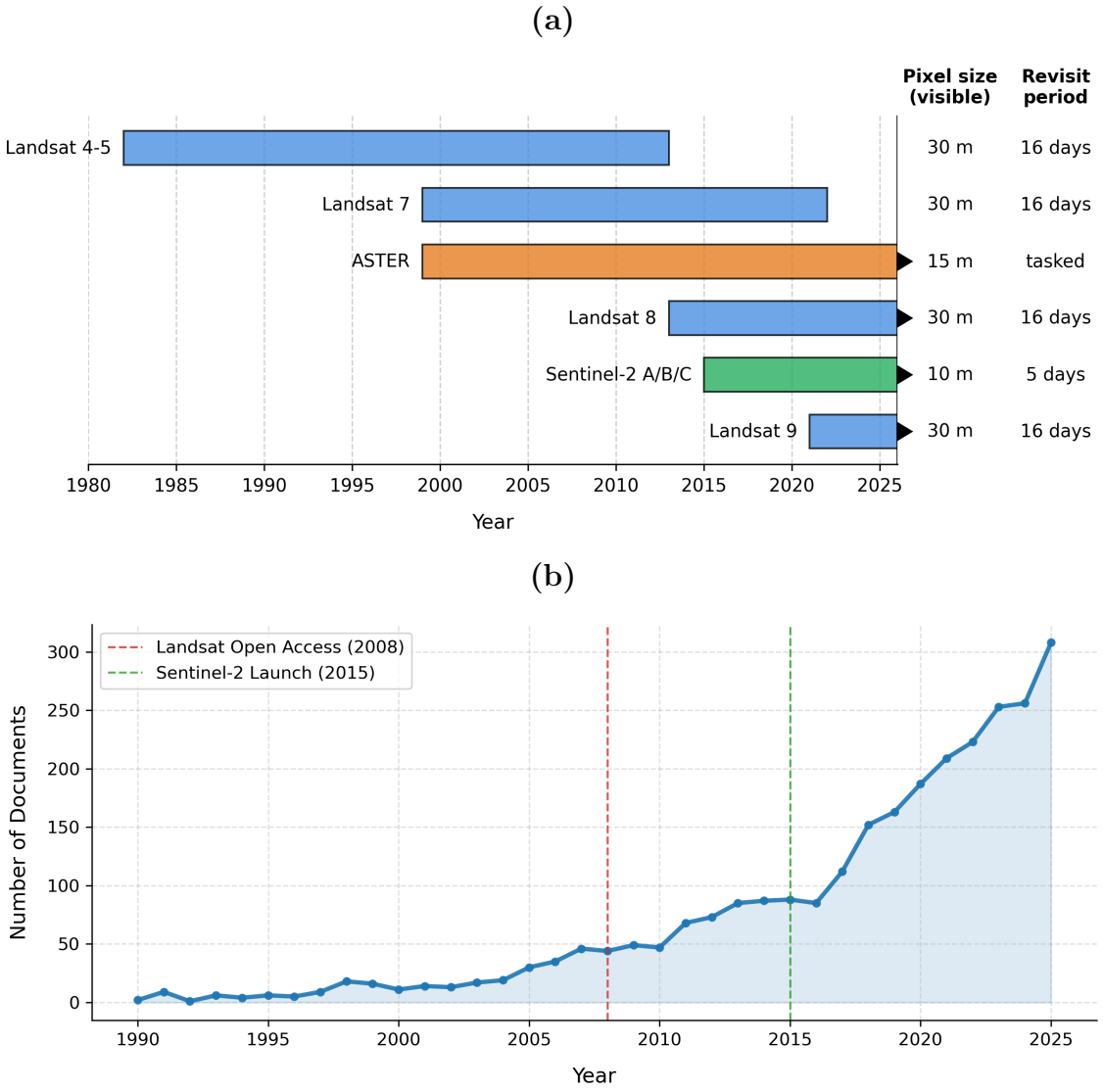


Figure 2.2: (a) Timeline for the publicly available satellites with global coverage. Source: Own elaboration, adapted and updated from Vos et al. (2019a). (b) Rapid evolution of the number of publications on satellite-derived shorelines (Number of documents per year for documents that include keywords "satellite" AND "shoreline" in Scopus database).

To leverage this massive and continuously expanding volume of "Big Data", the scientific community recognized that manual shoreline delineation constituted the bottleneck of the utilization of satellite imagery. This necessitated the development of automated computational frameworks designed to extract shoreline positions efficiently. Notable examples include CoastSat (Vos et al., 2019b), CASSIE (Almeida et al., 2021), SAET (Palomar-Vázquez et al., 2023), and SHOREX (Sánchez-García et al., 2020). While differing in their specific implementations, these tools generally share a common processing pipeline: automated image acquisition via cloud APIs (like Google Earth Engine), cloud and cloud-shadow masking, initial water/land segmentation at the pixel level using spectral indices, and finally, subpixel contouring techniques to delineate the final, continuous shoreline vector.

2.4 Traditional Waterline Extraction via Spectral Indices

The critical step in traditional SDW extraction pipelines is the initial segmentation of the image into "water" and "non-water" classes. This is almost universally achieved using Spectral Water Indices. These indices exploit the characteristic of water to absorb most radiation in the Near-Infrared (NIR) and Shortwave-Infrared (SWIR) wavelengths, while land (soil, sand, and vegetation) strongly reflects it (see Figure 2.3).

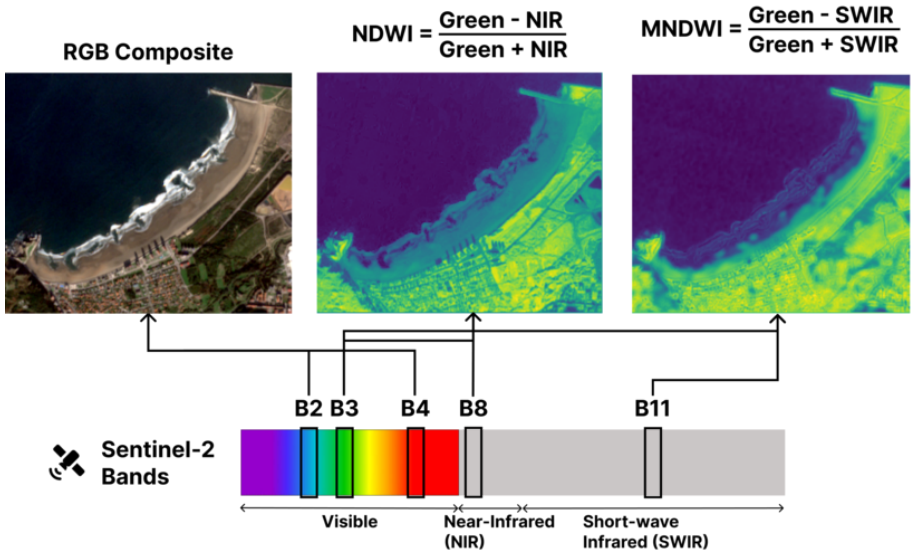


Figure 2.3: Schematic diagram of spectral index calculations using Sentinel-2 bands.

Historically, the Normalized Difference Water Index (NDWI), proposed by McFeeters (1996), and its variant, the Modified Normalized Difference Water Index (MNDWI) introduced by Xu (2006), have been the standard approaches for this task. Their implementation has been extensively documented in scientific literature to track long-term shoreline evolution and morphological changes across diverse coastal settings. For instance Ghosh et al. (2015), Liu et al. (2017) or Wang et al. (2017) demonstrated the efficacy of these indices in achieving high-accuracy shoreline detection using medium resolution satellite imagery, proving their value for multi-decadal coastal monitoring.

More recently, indices designed specifically to suppress environmental noise and improve performance in complex coastal scenes have been introduced. These include the Automated Water Extraction Index (AWEI), developed by Feyisa et al. (2014) to enhance water shadows and urban background removal, and the Subtractive Coastal Water Index (SCoWI) (Bergsma et al., 2024). The latter is particularly relevant to this project as it was developed specifically for Sentinel-2 imagery and has shown promising results in macrotidal environments.

Once the index is calculated to create a greyscale intensity map, a threshold must be applied to binarize the image. While setting a constant threshold (e.g., zero) is common, adaptive statistical methods like Otsu's thresholding (Otsu, 1979) are frequently used to find the optimal separation value based on the image's histogram distribution. After binarization, the shoreline is extracted by identifying the boundary between water and land pixels. This is typically done by finding the transition pixels between the two classes, which are then smoothed and converted into a vector line via subpixel contouring algorithms (such as marching-squares, Lorensen and Cline, 1987).

2.5 Challenges and Limitations in Macrotidal Environments

While traditional spectral indices combined with automated thresholding (e.g., Otsu thresholding) perform highly accurately (often achieving sub-pixel precision of <10 m) in microtidal, sheltered, or steeply sloped beaches, their reliability significantly degrades in complex, high-energy, and macrotidal environments. Recent global benchmarking efforts (Vos et al., 2023) have quantified this degradation, showing that horizontal extraction errors more than double (exceeding 20-40 m) on macrotidal beaches compared to their microtidal counterparts. Furthermore, site-specific assessments across the European Atlantic coast have similarly revealed profound limitations in these segmentation techniques during challenging oceanographic conditions (Cabezas-Rabadán et al., 2026).

The primary driver of error in these settings is the extensive and topographically complex intertidal zone exposed during ebb and low tides. In macrotidal beaches (tidal ranges > 4 m) characterized by intermediate or dissipative morphodynamic states, the retreating tide leaves behind vast plains of saturated sand. Spectrally, the transition from dry sand to wet sand to shallow water is not a sharp boundary but a gradual continuum. Indices such as MNDWI and AWEInsh consistently fail to differentiate between shallow water and saturated soil, systematically misclassifying the wet intertidal zone as "water" (see an example in Figure 2.4). This results in the extraction algorithms delineating the upper limit of the wet beach (or even the dune vegetation line) rather than the instantaneous water level, creating massive landward biases that can exceed hundreds of meters.

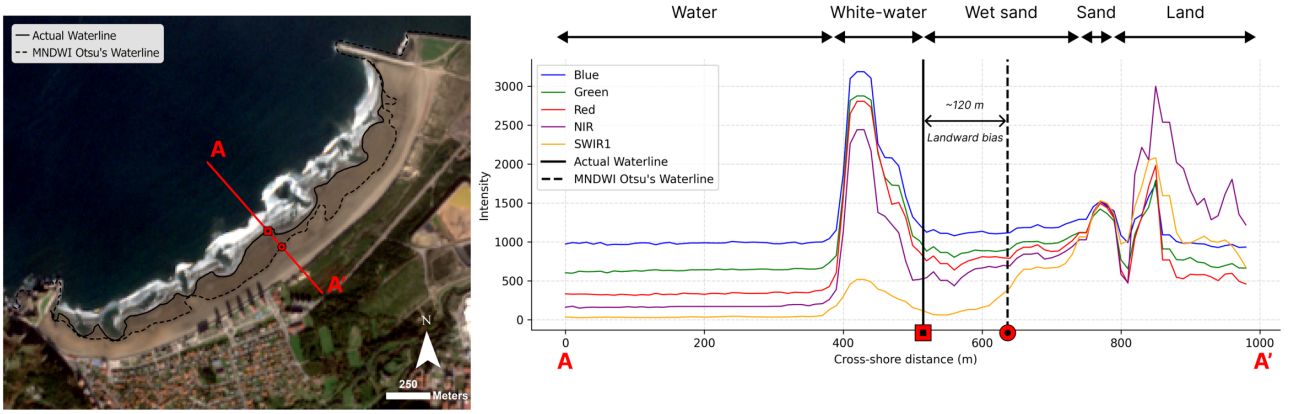


Figure 2.4: Cross-section of a macrotidal beach showing the spectral signature of water, white-water, wet sand, sand and land, derived from a Sentinel-2 image. The dashed line represents the shoreline extracted using MNDWI and Otsu's thresholding method, whereas the solid line represents the actual waterline. The MNDWI index is not able to differentiate between wet sand and water, leading to a landward bias in the extracted shoreline up to 120 m.

Furthermore, high-energy wave conditions may introduce additional errors. The presence of extensive white foam generated by breaking waves offshore drastically alters the expected low reflectance of the water surface. Indices like NDWI are particularly sensitive to foam, occasionally mistaking the outer breaker zone for land, thus producing severe seaward offsets (Hagenaars et al., 2018; Cabezas-Rabadán et al., 2026).

Ultimately, both specific case studies (Cabezas-Rabadán et al., 2026; Sánchez-García et al., 2020) and comprehensive benchmarking exercises (Vos et al., 2023) concur that the primary bottleneck for accurate SDW extraction in complex areas is not the geometric subpixel refinement algorithm, but the initial binary segmentation step (the spectral index determination and thresholding method) itself. Relying solely on the radiometric intensity of isolated pixels is insufficient to navigate the semantic ambiguity between wave foam, wet sand, and the actual waterline. This critical limitation strongly motivates the need to take a step back from classical spectral approaches and explore advanced, context-aware Deep Learning solutions.

2.6 The Shift towards Deep Learning in Earth Observation

To overcome the inherent limitations of pixel-by-pixel spectral thresholding, the remote sensing community is increasingly turning to Deep Learning (DL). Unlike traditional index thresholds that evaluate the reflectance value of an isolated pixel, advanced DL semantic segmentation architectures learn hierarchical spatial features by evaluating the spatial context, textures, and structural patterns. This contextual awareness allows models to distinguish the texture of wave foam from dry sand or recognize the spatial gradient of a saturated beach face, even when spectral signatures are nearly identical.

While the field is rapidly evolving, the technical landscape of DL for coastal monitoring is characterized by several competing paradigms. The encoder-decoder framework, exemplified by U-Net (Ronneberger et al., 2015) and its variants, remains the most prevalent architecture due to its effective use of skip connections to preserve fine spatial details (Tsiakos and Chalkias, 2023). More advanced convolutional models, such as DeepLabV3+ (Chen et al., 2018), have shown great potential by explicitly evaluating the image at multiple scales simultaneously. This wider field of view allows the model to better understand the broader coastal context (like the general geometry of the beach) while focusing sharply on the exact boundary, making it highly beneficial for extracting continuous, uninterrupted waterlines even in complex coasts.

Furthermore, a significant shift toward Vision Transformers (ViT) and hybrid architectures is underway. Models like SegFormer (Xie et al., 2021) and SETR (Zheng et al., 2020) demonstrate a superior ability to incorporate global contextual information, which is critical for distinguishing water boundaries in scenes affected by intense wave splash and structural shadows. More recently, dual-branch architectures such as TCCFNet (Fu et al., 2025) have emerged as highly synergistic networks, combining CNN layers for fine, local details with Transformer layers for large-scale scene understanding. Although TCCFNet was originally designed for mapping complex mangrove forests rather than strictly for shoreline extraction, its underlying mechanics are transferable to waterline mapping. By leveraging global context to understand the broader beach morphology while using local features to precisely trace the tricky land-water interface, this hybrid approach can drastically reduce the semantic confusion between visually similar elements like shallow water and wet sand. To push this focus on the transition zone even further, specialized boundary-aware networks (such as BS-Net (Jing et al., 2021) and BAWS (Zhan et al., 2025)) have been developed to treat the land-water interface as a primary feature rather than just the edge of a mask. By assigning dedicated interaction modules to explicitly model the boundary itself, these networks guarantee the high metric precision necessary for operational coastal monitoring.

Despite these architectural advancements, a critical analysis of the current state of the art reveals fundamental methodological gaps that this study aims to address:

- **Transition from Basic ML to Deep Contextual Learning:** While earlier AI applications for shoreline extraction heavily relied on traditional, pixel-wise Machine Learning classifiers (such as Support Vector Machines, Random Forests, or K-Nearest Neighbors), these methods still struggle with the complex spectral mixtures found in dynamic coastal zones (Tsiakos and Chalkias, 2023). Modern Deep Learning architectures overcome this by evaluating the spatial context rather than isolated pixels, though their full potential in adverse conditions is yet to be systematically quantified.
- **The Macrotidal Gap:** Most DL studies for shoreline extraction have been trained and validated in microtidal or sheltered environments where the land-water boundary is relatively stable. In macrotidal coasts, the "semantic ambiguity" between breaking wave foam, saturated sand, and shallow water remains a systematic challenge that most current models treat as noise rather than learned features (Tian et al., 2024). The effectiveness of DL models to navigate these specific, high-energy conditions is a critical void in current research.

- **Lack of Systematic Benchmarking vs. Operational Spectral Tools:** Although tools like CoastSat, CASSIE or SHOREX (which rely on spectral indices and automatic thresholding) are the current operational standard for automated shoreline extraction, there is a striking absence of extensive, consistent benchmarking that formally compares these standard pipelines against modern semantic segmentation models (Tian et al., 2024). Most DL studies evaluate new architectures against other neural networks rather than quantifying the exact magnitude of improvement over index-based methods in challenging environments.
- **Architectural Over-reliance and Class Complexity**: Current literature shows an over-reliance on U-Net variants without systematic justification for more complex coastal scenes (Tian et al., 2024). Furthermore, most models still approach the problem as a binary classification (water vs. land), failing to exploit the potential of multi-class segmentation (e.g., explicitly labeling "wet sand" or "white water") to reduce the confusion that leads to landward or seaward biases.

By presenting a direct, data-driven comparison of classical spectral thresholding and modern semantic segmentation models (specifically optimized for multi-class detection in macrotidal settings), this project directly targets these gaps. It seeks not only to improve accuracy but to provide a rigorous assessment of whether AI-based workflows offer a justifiable operational advantage over established index-based monitoring tools.

Chapter 3

Materials and Methods

This chapter details the comprehensive methodological framework developed to evaluate and benchmark the performance of traditional spectral methods against Deep Learning semantic segmentation models for macrotidal waterline extraction. To provide a clear roadmap of the research, Figure 3.1 illustrates the end-to-end processing pipeline. The workflow encompasses five primary stages: the definition of the study areas (Section 3.1); the acquisition and preprocessing of satellite and oceanographic data (Section 3.2); the generation of Ground Truth reference datasets (Section 3.3); the implementation of both traditional and AI-based extraction pipelines (Sections 3.4 to 3.6); and finally, the evaluation methodologies followed to assess the models (Section 3.7).

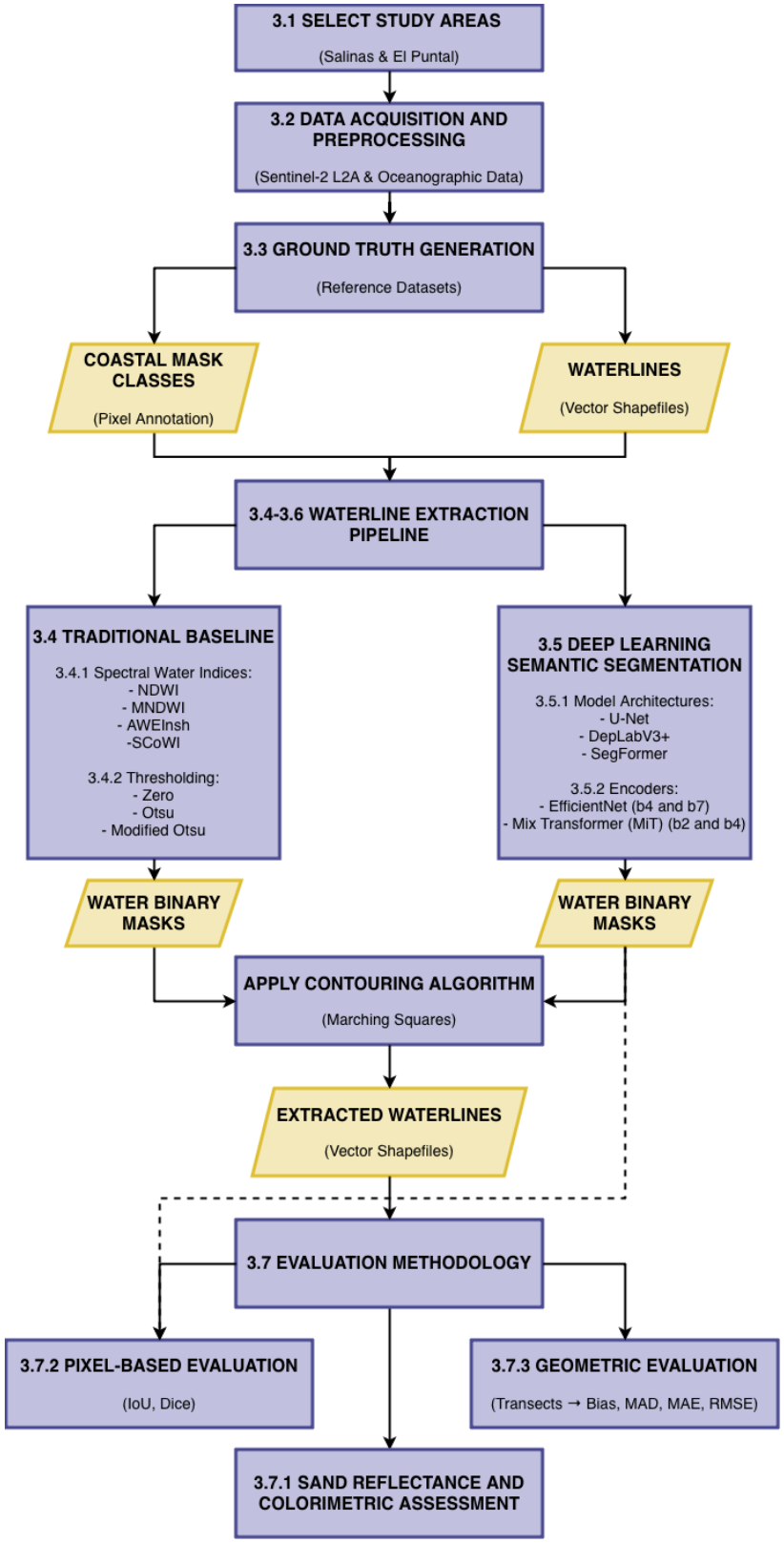


Figure 3.1: General methodological workflow of the study. The diagram outlines the data acquisition, model training, extraction pipelines, and evaluation stages detailed throughout this chapter.

3.1 Study Areas

The study focuses on two beaches located on the northern coast of Spain: Salinas (Asturias) and El Puntal (Cantabria) (Figure 3.2).

Salinas is a 2.5 km embayed sandy beach system located on the left bank of the Avilés estuarine inlet (Asturias). The beach is bounded by an inlet jetty to the east and a rocky headland (La Peñona) to the west. Its western sector, adjacent to the urban area of Salinas, is backed by a seawall and promenade, while the eastern sector (El Espartal) preserves a natural dune field rising up to 15 m above mean sea level. The beach is characterized by a wide intertidal zone (up to 300 m at spring tides) and a mean tidal range of 4.5 m, classifying it as a macrotidal environment. The wave climate is predominantly energetic and approach from the northwest, impacting directly with mean significant wave heights around 1.5 m and frequent storm events that can exceed 4.5 m. The sand often shows dark patches due to the accumulation of coal and carbonaceous shales, which can further complicate spectral differentiation between wet sand and water.

El Puntal is a 4 km sandy spit at the entrance of Santander Bay (Cantabria), formed by the interaction of estuarine and marine dynamics. The beach is backed by a dune system that reaches up to 10 m high and 50–100 m wide. The sandspit is located at its western end, whereas the eastern end is characterized by a more exposed morphology and the mouth of a small stream. In the same way as Salinas, El Puntal experiences a wide intertidal zone (up to 250 m) and a mean tidal range of around 4 m. However, unlike Salinas, El Puntal does not receive direct wave impact, as it is protected by the Magdalena Peninsula. Furthermore, also in contrast to Salinas, El Puntal features a much less dark, golden-colored sand.

The selection of Salinas and El Puntal as study areas responds to their representativeness of highly dynamic macrotidal environments, which consistently pose the most significant challenges for automated shoreline extraction methodologies. Both beaches share practically identical macrotidal ranges, featuring extensive intertidal zones that undergo profound daily fluctuations and result in the recurrent exposure of vast stretches of saturated sand. This inherently generates a critical spectral ambiguity between shallow water and wet sand, severely penalizing traditional pixel-by-pixel segmentation indices.

However, while they share this macrotidal nature, they exhibit key differences that allow for a comprehensive evaluation of the models under varying degrees of complexity. El Puntal, being sheltered from direct wave impact, serves as a baseline scenario with standard, golden-colored sand. Additionally, artificial dredging operations are routinely carried out at the very tip of El Puntal to maintain the required water depth for the navigation channel into the Bay of Santander. This dredging results in a much steeper beach slope in that specific area, which drastically reduces the extent of the intertidal zone, one of the main factors contributing to the reduced complexity of this environment. Summed to the other factors, this makes El Puntal a slightly less challenging environment compared to Salinas. In stark contrast, Salinas is fully exposed to energetic wave climates, resulting in a significantly higher presence of breaking wave foam (white-water) across a wide surf zone. Furthermore, the frequent accumulation of coal and carbonaceous shales along its intertidal profile creates noticeably dark patches of saturated sand. This stark color closely mimics the inherently low reflectance signature of water bodies in

optical satellite imagery. Together, the excessive foam and the deceptively dark wet sand render Salinas an ideal stress-test. It provides a robust framework to critically evaluate whether Deep Learning architectures can successfully leverage spatial and structural context to differentiate the actual waterline, overcoming the limitations where traditional spectral methods typically fail.

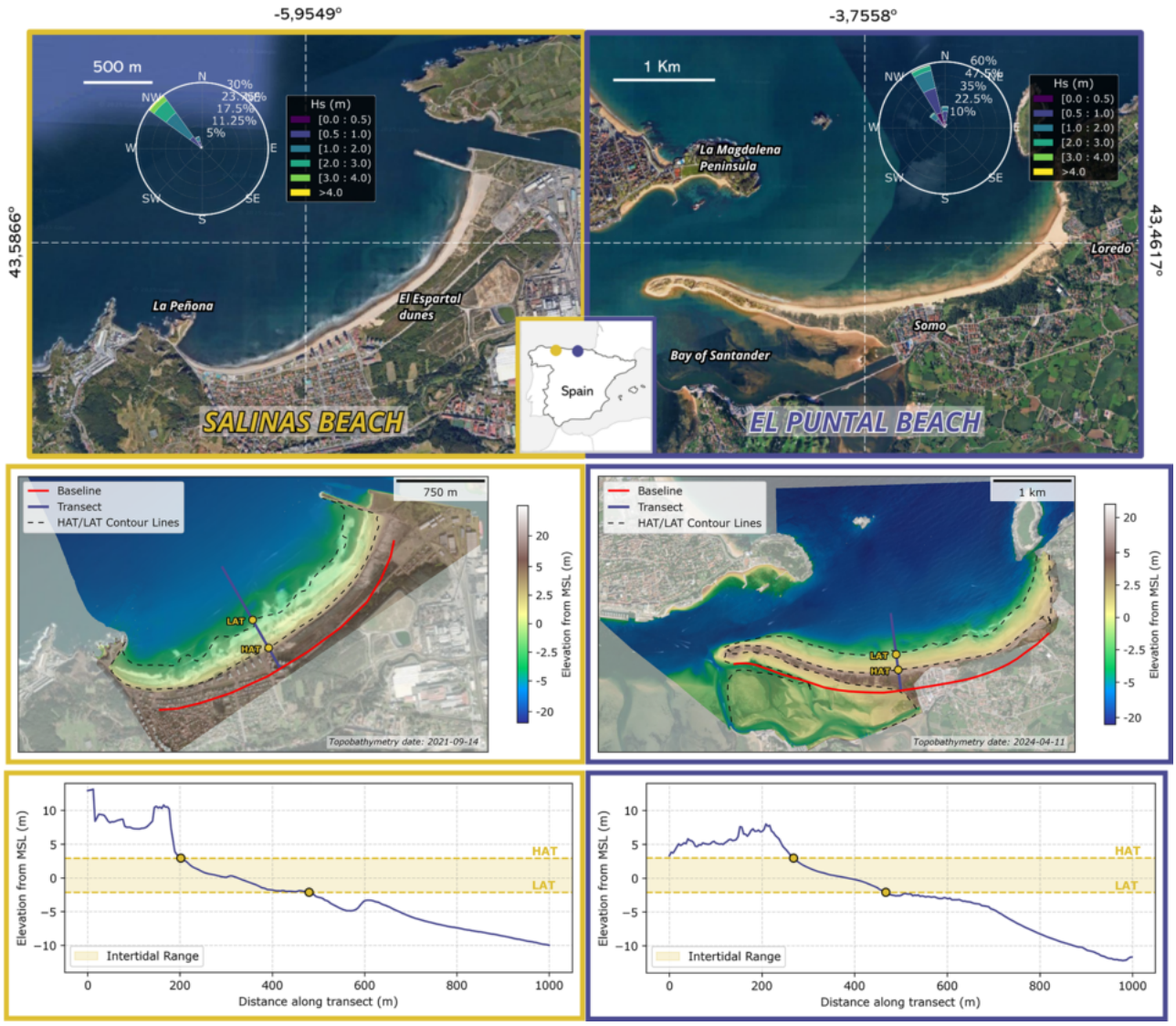


Figure 3.2: Location of the study sites: Salinas (left) and El Puntal (right) beaches. RGB image (top), topobathymetry (middle) using the ETRS89 / UTM 30N coordinate system, and beach profile (bottom) highlighting the intertidal range defined by the HAT/LAT (Highest/Lowest Astronomical Tide). (Source: Cabezas-Rabadán et al., 2026)

3.2 Data Acquisition and Preprocessing

An initial dataset of freely available Sentinel-2 Level-2A imagery, spanning the period from 2015 to 2025, was acquired for both study areas. The imagery was downloaded using the Python API for the Google Earth Engine. To ensure optimal visibility of the coastal features, an initial visual filtering process was applied to download only cloud-free scenes.

From the available multispectral data, six specific bands were selected to provide the necessary radiometric information for both the derivation of spectral indices and the feature extraction in Deep Learning models: Red (B04), Green (B03), Blue (B02), Near-Infrared (B08), Shortwave-Infrared 1 (B11), and Shortwave-Infrared 2 (B12).

Following the data acquisition, a dedicated preprocessing pipeline was implemented. First, the original massive satellite tiles were spatially cropped to strictly encompass the defined Regions of Interest (ROIs) for Salinas and El Puntal. Second, given that Sentinel-2 bands possess varying native spatial resolutions, spatial homogenization was required. The SWIR 1 and SWIR 2 bands, natively captured at a 20 m resolution, were resampled to match the 10 m spatial resolution of the visible and NIR bands. Finally, to prepare the dataset for the subsequent training phase of the Deep Learning semantic segmentation models (see Section 3.5), the preprocessed image ROIs were systematically divided into smaller patches of 128×128 pixels.

In addition to the optical imagery, historical oceanographic data provided by Puertos del Estado (https://www.puertos.es/) were acquired to accurately characterize the hydrodynamic conditions of each scene. Time-series records of sea level and wave climate were downloaded and temporally intersected to associate each Sentinel-2 image with its corresponding tide level and wave conditions at the exact moment of image acquisition. This comprehensive metadata association allows for the systematic evaluation and comparison of the extraction models' performance under varying oceanographic regimes. Furthermore, these associated tidal variables were directly utilized to perform a robust stratification of the dataset prior to the Deep Learning training phase, as detailed in Section 3.5.3.

However, it is crucial to acknowledge an inherent temporal sampling bias in the dataset stemming from the satellite's orbital mechanics. Sentinel-2 operates on a sun-synchronous orbit, ensuring it passes over the study sites at approximately the same local solar time (mid-morning). Because the local astronomical tide cycles are partially phase-locked with the solar day, this strict overpass schedule implies that the satellite systematically misses the absolute high waters of spring tides at this specific location. As illustrated in Figure 3.3, this results in a truncated tidal distribution where high tides are never captured. Conversely, the significant wave height (H_s) captured by the satellite generally preserves the overall statistical distribution of the full oceanic time-series, although extreme storm conditions remain underrepresented both due to their lower frequency and the fact that they are typically accompanied by heavy cloud cover, which filters them out during the optical image selection step.

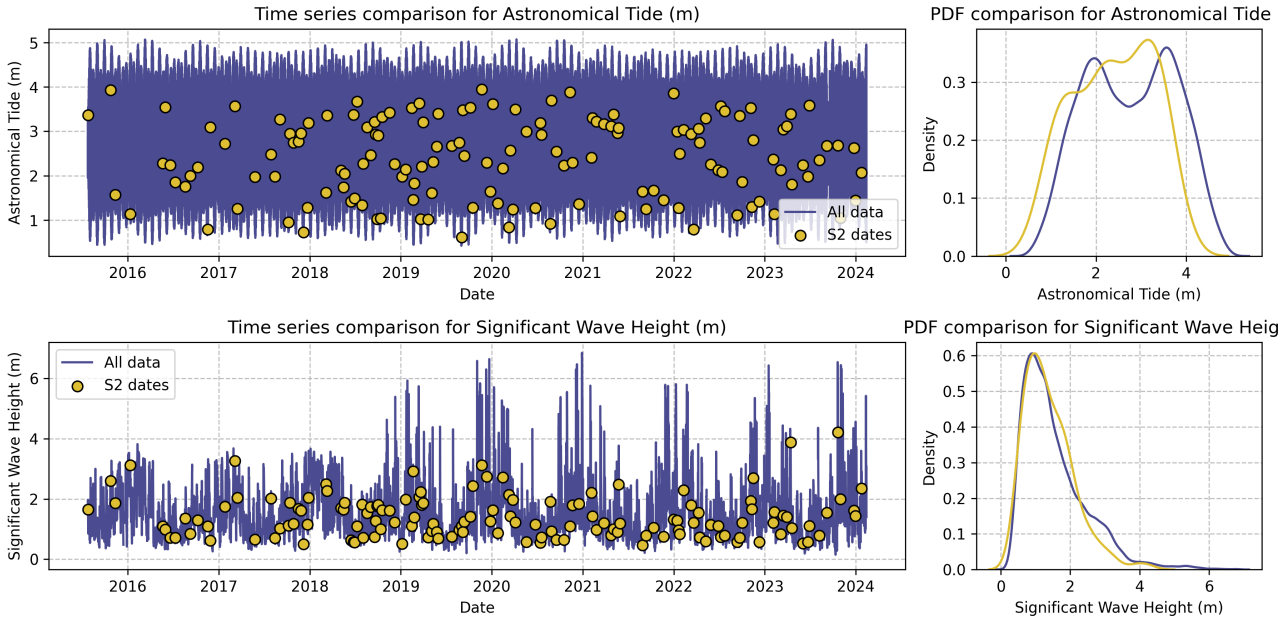


Figure 3.3: Comparison between the continuous oceanographic time-series (purple) and the conditions captured exactly during Sentinel-2 overpasses (yellow dots) for Salinas beach. The top panels highlight the Astronomical Tide, demonstrating the observational bias caused by the sun-synchronous orbit that systematically trims the high-tide peaks. The bottom panels show the Significant Wave Height (H_s), which more closely follows the theoretical distribution, albeit missing the extreme storm events.

3.3 Ground Truth Generation (Reference Datasets)

To rigorously train the algorithms and benchmark their performance, two distinct sets of Ground Truth (GT) data were generated: pixel-wise multi-class masks for the semantic segmentation models, and continuous vector lines for the final geometric benchmarking evaluation with spectral-index-based methods.

3.3.1 Coastal Mask Classes

To train and validate the Deep Learning models, a highly accurate pixel-level GT dataset was created. Given the complexity of the selected macrotidal environments, a multi-class approach was adopted rather than a simple binary (water/land) classification. The predefined classes were: *water* (calm water bodies), *white-water* (breaking wave foam in the surf zone), *wet sand* (the saturated intertidal zone), and *sand* (dry beach) (see Figure 3.4).

The annotation process was performed manually via photo-interpretation using the DigitalSreeni Image Annotator toolkit (Bhattiprolu, 2026) (Figure 3.4). To ensure the models focus solely on the relevant coastal features, any pixel not manually assigned to one of these four categories was

automatically designated as an *ignore_index*. This ensures that background elements (like urban infrastructure or vegetation) do not interfere with the loss function calculation during training. In total, a dataset comprising 161 image patches (combined from Salinas and El Puntal) was manually annotated and exported into the standardized COCO JSON format.

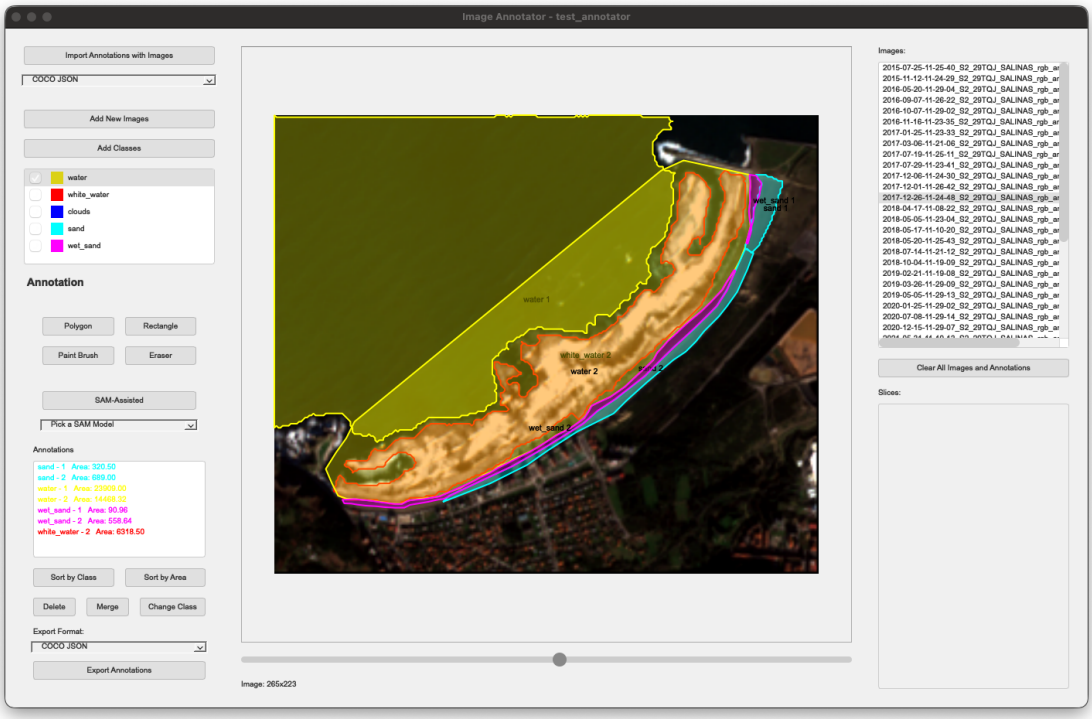


Figure 3.4: Screenshot of the DigitalSreeni Image Annotator software interface (Salinas beach). The overlaid mask displays the four annotated classes: water (yellow), white-water (red), wet sand (purple), and sand (cyan).

3.3.2 Waterline Vectorization

Alongside the pixel-wise masks, a separate reference dataset consisting of continuous vectorial waterlines was required to evaluate the geometric precision of the shorelines extracted by both the traditional indices and the DL models. For this purpose, a total of 285 reference waterlines (combined from Salinas and El Puntal) were generated.

This process was carried out using the geographic information system software ArcGIS Pro (Figure 3.5). The instantaneous waterline was manually digitized by tracing the spatial interface between the water mass and the emerged beach. The preprocessed 10 m resolution Sentinel-2 RGB composites were utilized as the visual background reference for this task. The resulting vector lines were then exported as standard Shapefiles (.shp), establishing the absolute geometric Ground Truth for the final benchmarking phase.

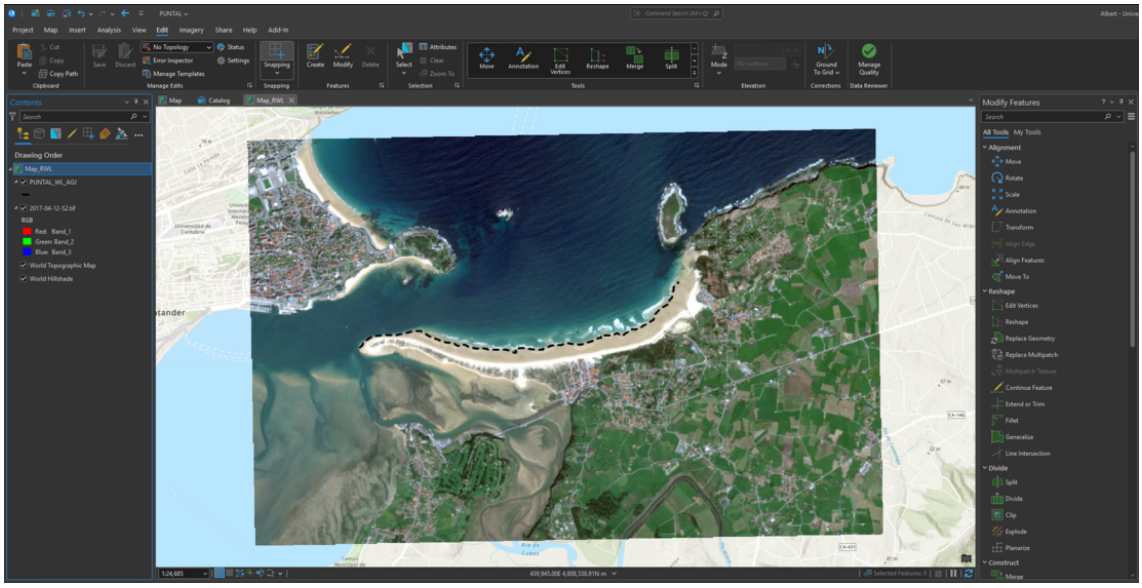


Figure 3.5: Screenshot of the ArcGIS Pro interface (El Puntal beach). The manually digitized waterline (black dashed line) is overlaid on the Sentinel-2 RGB composite.

3.4 Traditional Baseline: Spectral Indices and Thresholding

To establish a solid comparative baseline, we evaluated the performance of four different spectral water indices, coupled with three distinct thresholding techniques. This approach replicates the traditional operational pipelines currently dominating satellite-derived shoreline extraction.

3.4.1 Spectral Water Indices

Spectral indices are mathematical operations applied pixel-by-pixel across the different multispectral bands. By exploiting the distinct reflectance signatures of water and land, they heighten the contrast to facilitate water segmentation. We calculated four indices using the preprocessed Sentinel-2 bands: ρ_{Blue} (B02), ρ_{Green} (B03), ρ_{NIR} (B08), ρ_{SWIR1} (B11), and ρ_{SWIR2} (B12).

1. **Normalized Difference Water Index (NDWI):** Developed by McFeeters (1996), it maximizes the typical reflectance of water in the green light while minimizing its low reflectance in the near-infrared (NIR).

$$NDWI = \frac{\rho_{Green} - \rho_{NIR}}{\rho_{Green} + \rho_{NIR}} \tag{3.1}$$

2. **Modified Normalized Difference Water Index (MNDWI):** Proposed by Xu (2006) to address NDWI's difficulty in distinguishing water from built-up land. It replaces the NIR band with the Shortwave-Infrared (SWIR) band, which can penetrate shallow waters and wet environments slightly differently.

$$MNDWI = \frac{\rho_{Green} - \rho_{SWIR1}}{\rho_{Green} + \rho_{SWIR1}} \tag{3.2}$$

3. **Automated Water Extraction Index – No Shadows (AWEInsh):** Formulated by Feyisa et al. (2014) to maximize the separability of water from dark surfaces or topographic shadows. It combines multiple bands to suppress background noise.

$$AWEInsh = 4 \times (\rho_{Green} - \rho_{SWIR1}) - (0.25 \times \rho_{NIR} + 2.75 \times \rho_{SWIR2}) \tag{3.3}$$

4. **Subtractive Coastal Water Index (SCoWI):** Recently introduced by Bergsma et al. (2024) specifically for coastal applications with Sentinel-2 data. Unlike normalized ratios, it relies on a subtractive combination designed to better resolve dynamic intertidal zones and suppress the noise from wave breaking foam in macrotidal coasts.

$$SCoWI = \rho_{Blue} + 2 \times (\rho_{Green} - \rho_{NIR}) - 0.75 \times \rho_{SWIR1} - 0.5 \times \rho_{SWIR2} \tag{3.4}$$

3.4.2 Thresholding Techniques

Once the spectral indices are calculated, creating continuous unbouded greyscale intensity maps, a binarization process is required to extract the definitive land-water masks. This is achieved by applying a definitive cutoff threshold. To comprehensively benchmark this step, three different thresholding techniques were evaluated across all indices:

1. **Zero Thresholding:** A static approach fundamentally embedded in the original formulation of normalizated and subtractive water indices. Theoretically, water strongly absorbs NIR/SWIR light, rendering its ratio positive (> 0), while highly reflective land surfaces yield negative values (< 0). This method assumes a theoretical zero value as the universal boundary.

2. **Otsu's Thresholding:** A dynamic, image-adaptive algorithm widely established as the operational standard in modern extraction pipelines. Otsu (1979) method automatically determines the optimal threshold by calculating the value that minimizes the intra-class variance of the image's histogram under the assumption of a bimodal distribution (distinct water and land classes).

3. **Modified Otsu (Local Minimum Histogram):** Because standard Otsu thresholding can be mathematically skewed by disproportionate land/water class distributions or noisy elements unique to macrotidal beaches (like extensive wet sand patches dragging the histogram), we also evaluated a robust refinement proposed by Bergsma et al. (2024). In this modified approach, once the original Otsu threshold is calculated, the algorithm

defines a search interval between the two nearest local maxima on either side of the threshold. The definitive binarization cutoff is then set to the absolute local minimum of the histogram within that specific interval. This adjustment aims to physically anchor the threshold exactly in the valley between the two class distributions, ensuring a more context-aware segmentation.

3.5 Deep Learning Semantic Segmentation

To overcome the limitations of threshold-based spectral techniques, the core of this study evaluates the performance of state-of-the-art semantic segmentation neural networks. Similar to the baseline methodology, a systematic grid evaluation was designed by combining different model architectures with various feature-extracting encoders (backbones).

3.5.1 Model Architectures

Three widely recognized semantic segmentation architectures were selected to represent different evolutionary stages and paradigms in computer vision:

1. **U-Net:** A foundational encoder-decoder convolutional architecture originally developed for biomedical imaging (Ronneberger et al., 2015). Its defining feature is the use of skip connections that concatenate high-resolution feature maps from the encoder directly to the decoder. This enables the precise localization of fine details, which is highly advantageous for delineating a continuous, sharp waterline.

2. **DeepLabV3+:** An advanced Convolutional Neural Network (CNN) (Chen et al., 2018) that utilizes Atrous Spatial Pyramid Pooling (ASPP) to explicitly evaluate the image at multiple scales simultaneously. This architecture allows the model to capture a wider field of view and understand the broader coastal geometry without losing the resolution necessary to trace the complex land-water boundary accurately.

3. **SegFormer:** A novel and highly efficient Transformer-based architecture (Xie et al., 2021). Unlike traditional CNNs, it unifies a hierarchically structured Vision Transformer (ViT) encoder (which avoids strict positional encoding interpolation issues) with a lightweight multilayer perceptron (MLP) decoder. It excels at incorporating global contextual information, which is critical for navigating the semantic ambiguity of extensive wave splash and dark wet sand.

3.5.2 Encoders

The feature extraction capability of a segmentation network relies heavily on its encoder backbone. To balance computational efficiency and feature extraction depth, two distinct encoder families were chosen:

- **EfficientNet:** Representing the Convolutional Neural Network paradigm, we selected the *efficientnet-b4* (lighter, 17M params) and *efficientnet-b7* (heavier, 63M params) (Tan and Le, 2019). EfficientNet models are designed to optimize the trade-off between network depth, width, and resolution, making them robust for various image recognition tasks while maintaining computational efficiency.
- **Mix Transformer (MiT):** Representing the hierarchical Vision Transformer paradigm, we selected the *mit_b2* (lighter, 24M params) and *mit_b4* (heavier, 60M params) variants (Xie et al., 2021). These transformers are naturally designed to extract both local and global attention features.

It is worth noting a deliberate methodological divergence in this study's experimental matrix. From a strict theoretical standpoint, CNN-based decoders (such as U-Net or DeepLabV3+) are traditionally paired with CNN-based encoders, whereas Transformer decoders (like SegFormer) are paired with ViT encoders to maintain architectural purity. However, preliminary empirical testing during this project revealed an interesting dynamic: hybridizing the architectures (specifically, injecting the global-context awareness of the MiT transformer encoders into the precise local-feature CNN decoders) yielded remarkably superior and robust segmentation results in the most challenging coastal scenes. Consequently, to fully explore this advantage, an exhaustive grid approach was adopted where all candidate encoders (EfficientNet-B4, EfficientNet-B7, MiT-B2 and MiT-B4) were systematically crossed and tested across all three model architectures.

3.5.3 Training Setup

To ensure a fair and equitable comparison, all deep learning models were fine-tuned following an identical hyperparameter configuration and optimization strategy.

Prior to training, the annotated dataset was partitioned into a strict 80% training and 20% testing split, with a subset of the training data allocated for internal epoch validation. To prevent the models from overfitting to a specific tidal state or geographical morphology, this subdivision was strictly stratified based on both the associated instantaneous tide values (acquired during the preprocessing stage) and the specific study area (Salinas or El Puntal). This rigorous stratification ensures that images capturing high-tide and low-tide scenarios, as well as sheltered and exposed conditions, were equitably distributed. Consequently, the networks were trained and evaluated on a perfectly balanced representation of the highly dynamic intertidal zones.

The models were trained for a maximum of 100 epochs using a batch size of 16 patches. To minimize the loss function while mitigating overfitting, the AdamW optimizer (Adam with weight decay) was employed with an initial learning rate of 5×10^{-4} and a weight decay coefficient of 1×10^{-4}. Furthermore, a dynamic learning rate scheduler (*ReduceLROnPlateau*) was implemented, which reduces the learning rate by a factor of 0.5 if the validation loss plateaus for 5 consecutive epochs. Early stopping was also configured with a patience of 15 epochs to stop training when no further validation improvements were observed, thereby preventing over-parameterization and saving computational resources.

Given the multi-class formulation of the dataset, the standard Cross-Entropy Loss function was utilized. Crucially, the loss calculation explicitly ignored any pixel labeled as *ignore_index* (value 255) during the Ground Truth generation phase, forcing the networks to focus exclusively on learning the target coastal features (water, white-water, wet sand, and sand) without penalization from urban or vegetated background areas.

Finally, to significantly increase the model's robustness and generalization capabilities against varying beach orientations and viewing angles, a spatial data augmentation pipeline was strictly applied to the training set. Utilizing the Albumentations library, image patches were randomly subjected to horizontal flips, vertical flips, 90-degree rotations, and transpositions, each with a 50% probability during every training epoch. In contrast, the validation and test datasets were evaluated using the original, unaugmented distribution of the image patches.

To guarantee the statistical reliability of the results and account for the stochastic nature of network initialization and optimization, the training process for each model configuration (architecture-encoder pair) was repeated 10 independent times under varying random seeds. Following each set of iterations, the optimal run was selected based on its segmentation performance strictly along the coastal boundary zone. This was achieved by collapsing the multi-class predictions into binary masks (water versus non-water) and computing the Intersection over Union (IoU) exclusively within a reduced buffer region adjacent to the ground-truth waterline. A comprehensive explanation of this boundary-focused pixel-based evaluation and class mapping strategy is provided later in Section 3.7.2.

3.5.4 Hardware and Software Specifications

All computational routines throughout this project, including data preprocessing, vector operations, and statistical evaluation, were developed in Python. The functional source code containing the reproducible experimental pipelines has been open-sourced and published in a public GitHub repository (Gallego, 2026).

For the deep learning implementation, the *segmentation_models.pytorch* library was utilized as the primary API, providing structured and accessible implementations of the U-Net, DeepLabV3+, and SegFormer architectures. Rather than initializing the feature extraction encoders with random values, they were instantiated utilizing pre-trained weights from the *ImageNet* dataset (`encoder_weights="imagenet"`). This transfer learning strategy accelerates network convergence and provides the models with a pre-existing, robust understanding of fundamental geometric features such as edges and textures before fine-tuning them on the coastal domain.

Regarding hardware utilization, the computationally intensive task of training the semantic segmentation networks across all configurations, encoders, and random seeds was executed on a dedicated desktop PC equipped with an NVIDIA RTX 3060 GPU. Conversely, all other tasks (including dataset preprocessing, downloading of oceanographic data, programmatic thresholding of spectral indices, model inference, and the comprehensive spatial and geometric evaluations) were executed on an Apple MacBook Air featuring an M5 Apple Silicon chip.

3.6 Waterline Extraction Pipeline

To systematically compare the performance of traditional spectral indices against the Semantic Segmentation Deep Learning models, it is necessary to translate their respective pixel-based outputs into continuous, comparable geographic vectors. This process, known as sub-pixel contouring, is critical for achieving the high spatial precision required for shoreline monitoring.

For the traditional baselines (see Section 3.4), the extraction pipeline follows a strict three-step sequence (Figure 3.6). First, continuous intensity rasters are computed using the mathematical formulations of the four selected spectral water indices (i.e., NDWI, MNDWI, AWEInsh and SCoWI). Second, discrete binary masks ("water" versus "non-water") are created by applying the cutoff threshold values evaluated for each method (Zero, Otsu, and Modified Otsu). Third, the final waterlines are generated in vector format by utilizing the Marching Squares contouring algorithm (Lorensen and Cline, 1987) to trace a smooth boundary where the binary mask transitions from 0 to 1. This algorithm effectively interpolates the exact sub-pixel location of the water boundary, allowing for a more precise vector representation of the waterline.

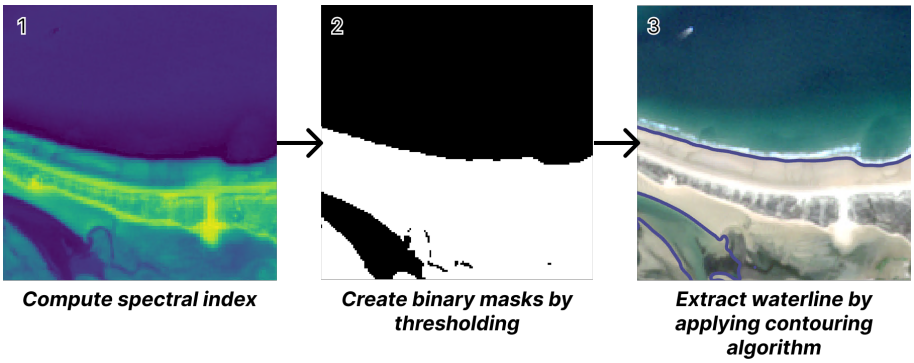


Figure 3.6: Diagram illustrating the waterline extraction pipeline for traditional spectral indices. The process involves a three-step sequence: (1) calculation of the continuous spectral index raster, (2) application of the cutoff threshold to create a binary mask, and (3) extraction of the waterline vector using the Marching Squares contouring algorithm.

Conversely, for the Deep Learning models, the segmentation output requires a slight mathematical adaptation to maximize its utility for vectorization. After processing a 128×128 image patch, the neural network outputs multi-class prediction logits. These logits are converted into confidence scores (ranging from 0.0 to 1.0) using a Softmax activation function. Subsequently, the pixels segmented as *water* and *white-water* classes are collapsed into a single "water" class, while the *wet sand* and *sand* classes are collapsed into a single "non-water" class. Then, the final waterlines are extracted using the same Marching Squares contouring algorithm as in the traditional pipeline, applied to the resulting binary mask. This ensures that the final output from both approaches is directly comparable in terms of format and spatial precision.

Finally, for both traditional and DL approaches, the individual continuous line segments extracted in each image are merged together. A morphological length filter is then applied, discarding any discontinuous segments shorter than a minimum threshold length. This filtering effectively removes small landlocked ponds, temporary tidal puddles, and isolated false positive artifacts, resulting in a single, robust, and continuous vector representation of the waterline.

3.7 Evaluation Methodology

To comprehensively assess the performance of the tested methodologies, the accuracy of the outputs was evaluated across two distinct structural domains: a purely pixel-based evaluation of the semantic segmentation masks, and a rigorous geometric validation of the final continuous vector waterlines. In addition to these core evaluations, an independent empirical analysis of the sand reflectance and colorimetry was conducted to physically contextualize the spectral challenges inherent to the study sites, particularly in Salinas. This analysis is crucial to understand the underlying physical factors that may influence the performance of both traditional and AI-based extraction methods.

3.7.1 Sand Reflectance and Colorimetric Assessment

Prior to the core pixel-based and geometric evaluations, an independent empirical analysis was conducted to quantify the spectral differences of the supratidal sand between the two study sites. This assessment was designed to test the hypothesis that the severe degradation in the segmentation performance of traditional spectral indices at Salinas is fundamentally driven by the physical coloration of its beach. Specifically, the accumulation of dark carbonaceous shales and coal creates a low-reflectance environment that critically mimics the spectral signature of water, in stark contrast to the highly reflective, golden-colored sand of El Puntal.

To quantitatively verify this spatial divergence without manual intervention across the entire temporal dataset, a strict constraint-based methodology was implemented to isolate the spectral signature of dry sand. First, a single "candidate image" was carefully selected for each beach. These reference images were chosen based on optimal conditions for dry beach exposure: cloud-free weather, minimum astronomical tide, and low energetic wave conditions. Utilizing these candidate scenes, a static Region of Interest (ROI) polygon was manually digitized strictly over the emerged, dry sand area of each beach.

To extract a reliable time-series of sand reflectance values, it was imperative to ensure that the static ROIs were not contaminated by water or wave run-up in subsequent images. Therefore, the complete dataset was temporally filtered using the historical oceanographic metadata. For each beach, only the satellite acquisitions where the instantaneous tidal level, significant wave height (H_s), and peak period (T_p) were equal to or strictly lower than those of their respective candidate image were retained. This rigorous hydrodynamic thresholding guarantees that the pixels contained within the static ROIs predominantly represent undisturbed dry sand throughout the evaluated subset.

Finally, using the preprocessed Sentinel-2 Level-2A imagery, two specific physical metrics were calculated for all pixels within the isolated ROIs to characterize the brightness and colorimetry of the sand:

1. **Mean Visible Reflectance ($\bar{\rho}_{RGB}$):** Calculated as the unweighted arithmetic mean of the Red, Green, and Blue bands. This metric serves as a direct proxy for the overall albedo of the sand in the visible spectrum, evaluating its absolute darkness.

$$\bar{\rho}_{RGB} = \frac{\rho_{Red} + \rho_{Green} + \rho_{Blue}}{3} \tag{3.5}$$

2. **Red-Blue Differential (Δ_{R-B}):** This metric evaluates the colorimetric slope of the target. Standard golden sand typically exhibits strong reflectance in the red spectrum and higher absorption in the blue, yielding a large positive differential. Conversely, carbonaceous dark sand presents a flatter, wavelength-independent spectral response (grey/black), resulting in a significantly lower differential.

$$\Delta_{R-B} = \rho_{Red} - \rho_{Blue} \tag{3.6}$$

By comparing the probabilistic distributions of these two metrics between Salinas and El Puntal, this analysis establishes a measurable geologic and spectral baseline. This baseline is crucial to physically contextualize the statistical geometric errors evaluated in the subsequent algorithmic sections.

3.7.2 Pixel-based Evaluation

Evaluating segmentation performance across full coastal scenes introduces a significant statistical bias due to extreme spatial class imbalance. When traditional pixel-based metrics are calculated over the entire image grid, their scores are highly susceptible to artificial inflation. Because the vast majority of pixels in a standard satellite patch belong to unambiguous, easily classified regions (such as the open ocean or stable inland areas) a model can achieve exceptionally high scores even if it completely misidentifies the delicate land-water boundary. Since the actual intertidal transition zone constitutes only a minuscule fraction of the total image, critical segmentation errors along the shoreline are mathematically overshadowed by the large number of correctly classified background pixels.

To counteract this statistical bias, a highly constrained, boundary-focused evaluation protocol was designed. Initially, the native multi-class structure of the ground-truth masks (and the corresponding Deep Learning predictions) was mathematically collapsed into binary domains: *water* and *white-water* were unified under the definitive "Water" class, while *wet sand* and *sand* were clustered into "Land". Any unannotated background elements assigned with the *ignore_index* (value 255) were strictly excluded from the calculations.

Subsequently, a symmetric spatial buffer equivalent to 10 pixels (representing a physical distance of 100 meters, given Sentinel-2's native 10 m resolution) was dynamically generated exclusively around the exact ground-truth waterline (see Figure 3.7). The evaluation was strictly constrained to this 100-meter buffer zone algorithmically identified using the Euclidean Distance Transform. By completely masking out the deep water body and the inland sectors, the models were forced to perform, and be penalized, exclusively on the challenging semantic ambiguity of the intertidal zone.

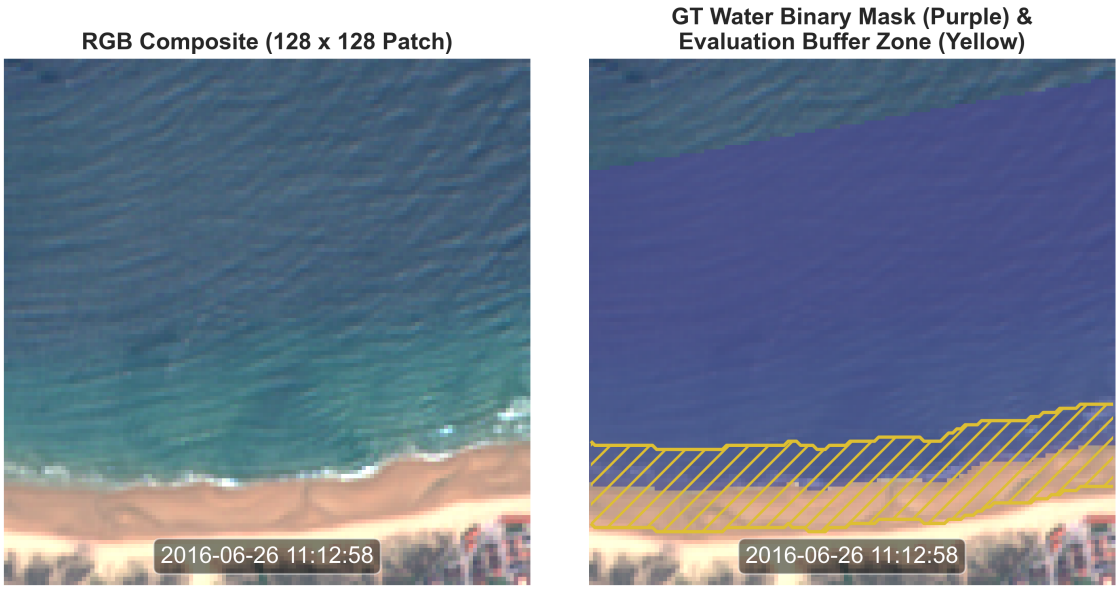


Figure 3.7: Example of the pixel-based evaluation mask. The left panel shows the original RGB image. The dynamic evaluation buffer (right) highlights the 10-pixel zone around the GT waterline where the pixel-based metrics are calculated, effectively excluding the deep water and inland areas from the evaluation.

Within this restricted boundary zone, the segmentation performance was quantified using two standard computer vision metrics:

- **Intersection over Union (IoU):** Also known as the Jaccard Index, it calculates the ratio of the intersection of the predicted and ground-truth water masks to their union. It is a strict metric that heavily penalizes false positives (e.g., misclassifying wet sand as water).
- **Dice Coefficient:** Similar to the IoU but assigning double weight to the correct positive predictions, providing a slightly more forgiving, harmonized measure of the overlap.

3.7.3 Geometric Evaluation

While the pixel-based evaluation determines how well a model categorizes the image grid, the geometric evaluation translates those results into the real-world metric required by coastal engineers: tracking the physical position of the coast in meters.

To achieve this, the continuous vectorial waterlines derived from both the traditional sub-pixel contouring and the Deep Learning probability maps (as described in Section 3.6) were benchmarked against the manually digitized Ground Truth shapefiles. This assessment was performed using the standard shore-perpendicular cross-shore transect methodology (Figure 3.8).

First, a fixed, static landward baseline roughly parallel to the general orientation of the beach is established. From this baseline, perpendicular transects were cast seaward at evenly spaced intervals (i.e., every 50 meters) along the entire longitude of the study area. For each individual transect (i), the intersection points with both the predicted vector waterline and the reference GT waterline were mathematically calculated. The 1D Euclidean distance from the baseline to these two points constitutes the basis for the extraction error at that specific profile. It's worth mentioning that, within the specific context of remote sensing applied to waterline monitoring, this calculated error (the spatial distance between the actual and the predicted position) is widely referred to in the literature as the extraction "offset". Mathematically, the offset for a specific transect i is formulated as:

$$Offset_i = |d_{mod,i} - d_{obs,i}| \tag{3.7}$$

where $d_{mod,i}$ represents the cross-shore distance from the baseline to the *modeled* (predicted) shoreline derived from either the DL or index-based methods at transect i, and $d_{obs,i}$ represents the distance to the *observed* reference shoreline extracted from the Ground Truth.

By aggregating the distances recorded across all intersecting transects for a given satellite image, several robust statistical metrics were computed:

- **Median Absolute Deviation/Error (MAD):** The median of the absolute offsets. Because it relies on the median rather than the mean, MAD is highly robust against severe outliers (such as specific transects with an extreme errors due to a contouring artifact or specific complex scenarios because a highly energetic wave event), offering a highly representative value of the "typical" coastline error.
- **Interquartile Range (IQR):** The range between the 25th and 75th percentiles of the offsets. This metric measures statistical dispersion.
- **Mean Absolute Error (MAE):** The average of the absolute values of the offsets. Unlike Bias, MAE does not allow positive and negative errors to cancel each other out, providing a straightforward measure of the average magnitude of the extraction error.
- **Mean Error (Bias):** The arithmetic average of all transect offsets. It evaluates the systematic tendency of the model to underestimate or overestimate the water extent. A

negative Bias indicates a systematic landward offset (the model predicts water further inland than reality, a common failure over wet sand), while a positive Bias indicates a seaward bias.

- **Root Mean Square Error (RMSE):** Measures the overall magnitude of the spatial discrepancy. Because it squares the errors before averaging, the RMSE strongly penalizes extreme localized deviations (such as the model mistakenly tracing an isolated intertidal pool or wave foam), providing a conservative and reliable metric of the extraction's operational viability.
- **Coefficient of Determination (R^2):** Evaluates the degree of correlation between the predicted and observed shoreline positions across all transects. An R^2 value close to 1 indicates a strong linear relationship, while a value close to 0 suggests no correlation.

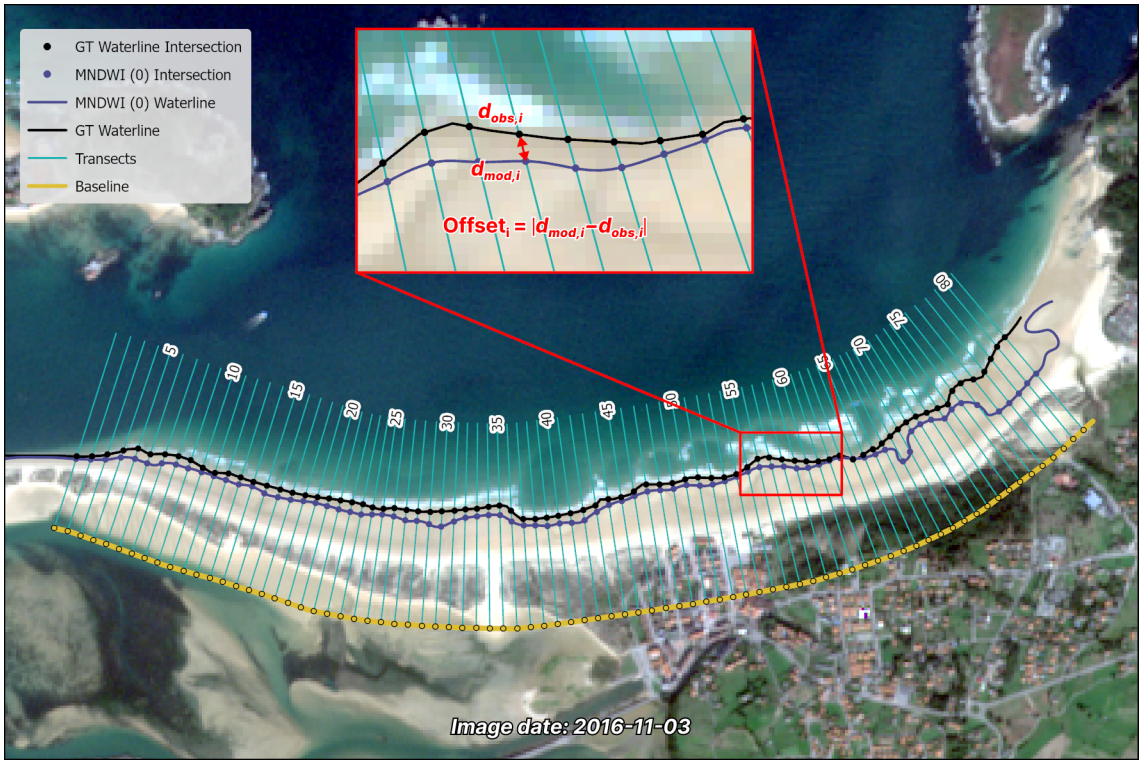


Figure 3.8: Example of the perpendicular transect methodology for geometric evaluation. The black line represents the manually digitized GT waterline, while the purple line represents the predicted waterline from either the DL or index-based method (in this case, MNDWI using Zero threshold). The yellow line is the fixed landward baseline. The cyan lines are the perpendicular transects cast from the baseline to the predicted and GT waterlines. The offset is calculated as the absolute distance between the intersection points of the predicted and GT waterlines along each transect.

Statistical Analysis Framework:

Once the aforementioned cross-shore metrics were derived for the entire dataset, a comprehensive statistical framework was established to interpret the results and facilitate a structured comparison in Chapter 4. This framework relies on several analytical approaches to evaluate the models across different dimensions:

- **Global and Distributional Analysis:** To compare the overarching performance between Index-based and Deep Learning families, as well as to identify morphological dependencies, the aggregated offset distributions are analyzed using boxplots and violin plots segregated by study area.
- **Spatial Bias Evaluation:** Beyond absolute error magnitudes (MAE), the systematic orientation of the errors is evaluated. Hexbin scatter plots are utilized to map the extraction bias (landward vs. seaward) across the cross-shore distances, revealing intrinsic behaviors of specific architectures and thresholding techniques.
- **Spatial and Environmental Dependency Assessment:** To isolate the sources of error variability, the extraction offsets are both evaluated spatially (transect-by-transect) and cross-referenced with hydrodynamic forcing. Specifically, the offsets are analyzed across categorical quartiles of tidal levels and significant wave heights (H_s) to assess model degradation under complex coastal dynamics.
- **Cross-Site Generalization:** The robustness of the models is quantified by calculating the MAE difference (Δ MAE) between the sheltered baseline site (El Puntal) and the highly dynamic stress-test site (Salinas). A minimized Δ MAE serves as the primary indicator of a model's capacity to generalize without overfitting to local morphologies.
- **Multi-Criteria Ranking:** Finally, all computed metrics (RMSE, MAE, MAD, Bias, and R^2) are synthesized and normalized into heatmaps and bump charts to produce a definitive, objective ranking of the top-performing global configurations.

3.7.4 Summary of Evaluated Models

To provide a clear overview of the entire experimental matrix, Table 3.1 summarizes all the models and configurations tested in this project, explicitly detailing which evaluation protocols were applied to each.

Table 3.1: Summary of the evaluated extraction methods, their configurations, and the corresponding applied evaluation protocols.

Category	Base Model / Index	Evaluated Variations / Encoders	Pixel-based Eval.	Geometric Eval.
Index-based	NDWI	Zero, Otsu, Modified Otsu	No	Yes
	MNDWI	Zero, Otsu, Modified Otsu	No	Yes
	AWEInsh	Zero, Otsu, Modified Otsu	No	Yes
	SCoWI	Zero, Otsu, Modified Otsu	No	Yes
Deep Learning-based	U-Net	EfficientNet-B4/B7, MiT-B2/B4	Yes	Yes
	DeepLabV3+	EfficientNet-B4/B7, MiT-B2/B4	Yes	Yes
	SegFormer	EfficientNet-B4/B7, MiT-B2/B4	Yes	Yes

Chapter 4

Results

4.1 Physical Context: Sand Reflectance and Colorimetric Assessment

To establish the physical baseline of the study areas and test the hypothesis regarding the spectral ambiguity at Salinas beach, the brightness and colorimetry of the supratidal sand were quantified. Following the strict hydrodynamic filtering detailed in Section 3.7.1, a total of 28 optimal scenes for El Puntal and 24 scenes for Salinas were retained for this analysis. Figure 4.1 illustrates the probabilistic distributions of the two evaluated metrics across both locations.

The Mean Visible Reflectance ($\bar{\rho}_{RGB}$) demonstrates a significant disparity in the overall albedo between the two sites. El Puntal exhibits highly reflective sand with a median intensity value of 2293.15. In stark contrast, the sand at Salinas is considerably darker, yielding a median reflectance of 1737.90. In relative terms, this represents a difference of -25% for Salinas. This absolute reduction in baseline brightness physically confirms the presence of highly absorbent carbonaceous materials mixed within the Salinas intertidal zone.

Furthermore, the Red-Blue Differential (Δ_{R-B}) reveals an even more pronounced divergence in the colorimetric signature of the beaches. El Puntal displays the classic physical signature of golden sand (reflecting significantly more red light than blue) resulting in a high median differential of 963.45. Conversely, Salinas presents a drastically flattened spectral response, with a median differential of 206.23. In relative terms, this represents a difference of -80% for Salinas. As observed in the boxplots, the interquartile ranges of these distributions do not overlap, indicating a fundamentally different geological coloration. The exceptionally low Δ_{R-B} at Salinas quantitatively proves that its sand behaves spectrally more like a neutral grey or black body, rather than standard beach sand.

These empirical findings successfully validate the classification of Salinas as a morphological stress-test environment. The inherently dark and spectrally flat nature of its sand creates a low-reflectance background that closely mimics the radiometric signature typical of water bodies in optical imagery. This structural baseline establishes a critical semantic ambiguity that is hypothesized to severely penalize the segmentation performance of traditional normalized spectral indices, a phenomenon that will be systematically addressed in the subsequent geometric evaluations (see Section 4.3).

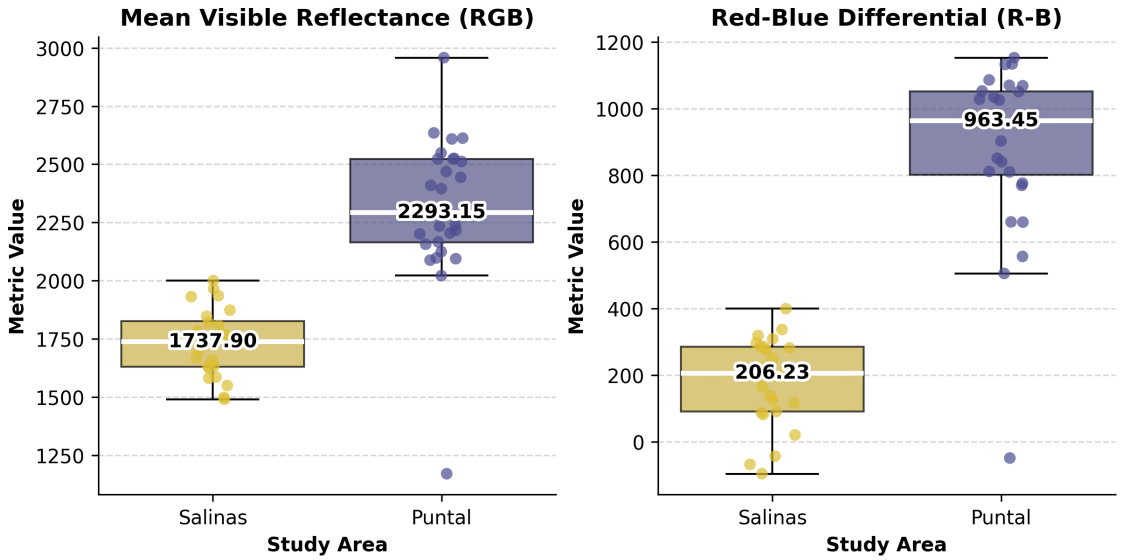


Figure 4.1: Comparison of the sand reflectance metrics between El Puntal and Salinas. The left panel displays the absolute albedo proxy ($\bar{\rho}_{RGB}$), while the right panel highlights the colorimetric slope (Δ_{R-B}). Boxplots represent the interquartile range and median, with overlaid stripplots detailing individual scene observations.

4.2 Deep Learning Pixel-based Performance

Prior to evaluating the geometric accuracy of the extracted shorelines (Section 4.3), it is imperative to assess the internal semantic representations learned by the Deep Learning architectures. By analyzing the pixel-by-pixel classification performance, the models' capacity to correctly interpret the complex physical elements of the coastal environment can be validated. This evaluation initially focuses on the multi-class segmentation task across four distinct coastal categories: Water, Sand, Wet Sand, and White Water.

The median Intersection over Union (IoU) and Dice coefficient metrics reveal a robust baseline understanding of the macroscopic coastal environments across both study areas. The models excel at identifying the primary definitive features, achieving outstanding median IoU scores for the *Water* class (0.981 at El Puntal, 0.928 at Salinas) and the dry *Sand* class (0.914 at El Puntal, 0.832 at Salinas).

Crucially for macrotidal applications, the semantic isolation of the *Wet Sand* class proved highly successful. The networks maintained a median IoU of 0.875 and 0.843 (Dice: 0.933 and 0.915) at El Puntal and Salinas, respectively. This strong performance mathematically justifies the geometric superiority of the Deep Learning models observed in later sections; the architectures are explicitly capable of differentiating the water-saturated intertidal zone from the actual instantaneous waterline, avoiding the severe radiometric confusion that critically penalizes traditional spectral indices.

The most notable variance in the multi-class assessment occurs within the *White Water* category, which highlights the direct influence of local hydrodynamic conditions on model training and inference. At the exposed, high-energy beach of Salinas, the models achieve a functional median IoU of 0.651 (Dice: 0.788), successfully delineating the prominent surf zone. Conversely, at El Puntal, the *White Water* performance collapses to a median IoU of 0.159 (Dice: 0.274). Rather than an architectural failure, this is a direct reflection of the physical environment: El Puntal is characterized with a lower frequency wave foam and a narrower extent of white water features. This severe class imbalance natively limits the predictive confidence of the model in that specific region, though it bears no negative impact on the overall shoreline extraction since the feature itself is absent.

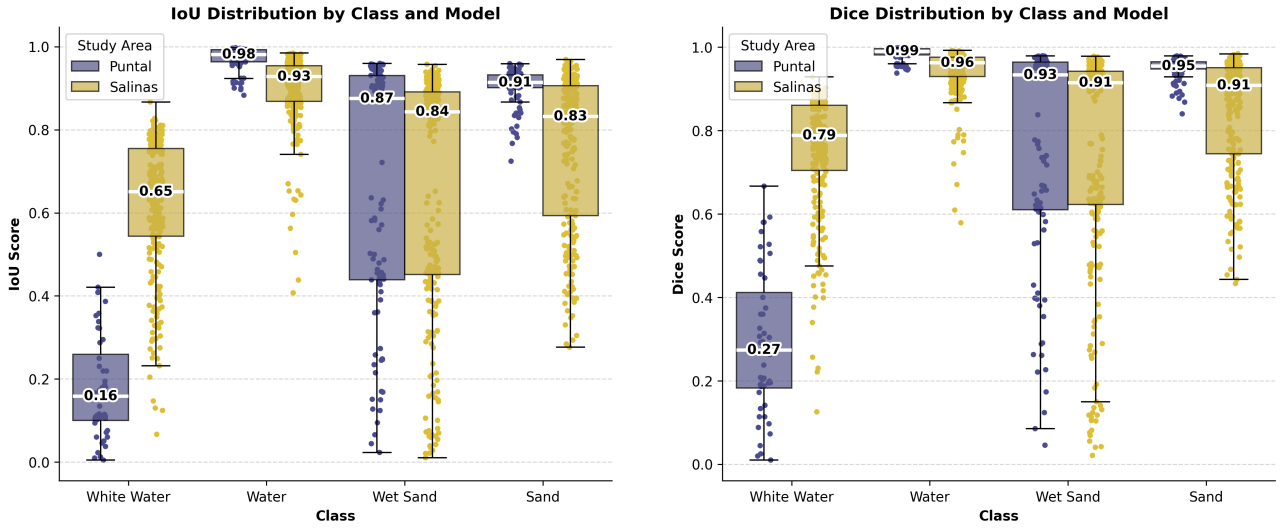


Figure 4.2: Pixel-based performance distributions (IoU and Dice coefficient) for the four annotated coastal classes, disaggregated by study area. The networks demonstrate exceptional accuracy for standard macro-features (Water, Sand) and maintain highly robust predictions for the morphodynamically complex Wet Sand. The performance drop in White Water at El Puntal is directly attributed to the physical scarcity of wave foam in the sheltered estuarine environment.

To align the pixel-based evaluation with the ultimate objective of the study (the extraction of a singular geometric waterline) the four multi-class categories were collapsed into a binary classification problem: *Water* (incorporating Water and White Water) versus *Non-water* (incorporating Sand and Wet Sand).

When evaluated under this binary paradigm, the semantic performance of the Deep Learning models reaches near-saturation. As illustrated in the right panel of Figure 4.3, all evaluated architectures achieved an exceptional Mean IoU tightly bounded between 0.924 and 0.941. On a standard absolute scale, the visual variance among the models is virtually imperceptible. This confirms that the fundamental semantic delineation of the land-water boundary is definitively resolved by modern convolutional and transformer-based architectures, effectively eliminating the radiometric confusion that plagues traditional indices.

Because absolute semantic accuracy is saturated, the critical differentiator among the models shifts from raw predictive power to computational efficiency and architectural optimization. The left panel of Figure 4.3 maps the Mean IoU against the total number of trainable parameters for each configuration.

Counterintuitively, the highest performance is consistently achieved by the lighter architectures. The absolute maximum binary accuracy was recorded by SegFormer (MiT-B2), yielding a 0.941 mean IoU with a highly efficient footprint of approximately 24.7 million parameters. Similarly, the U-Net (MiT-B2) secured a 0.936 mean IoU requiring only 27.5 million parameters.

In stark contrast, scaling up the network capacity resulted in a marginal, yet systematic, degradation of performance. All configurations utilizing the heavy EfficientNet-B7 or MiT-B4 encoders (which demand computational footprints ranging from 61.3 to 67.1 million parameters) scored below the 0.935 mean IoU threshold. This inverse relationship between parameter count and binary accuracy highlights a critical architectural insight for the highly specific task of shoreline delineation: overparameterization does not yield better semantic abstraction. Instead, excessively deep encoders likely induce a degree of spatial overfitting, where the network sacrifices the fine-grained, high-resolution localization required to trace a sharp boundary in favor of unnecessary complex feature extraction. This pixel-level degradation of the boundary sharpness is a fundamental phenomenon that will be explicitly corroborated during the subsequent geometric offset evaluation (Section 4.3.2).

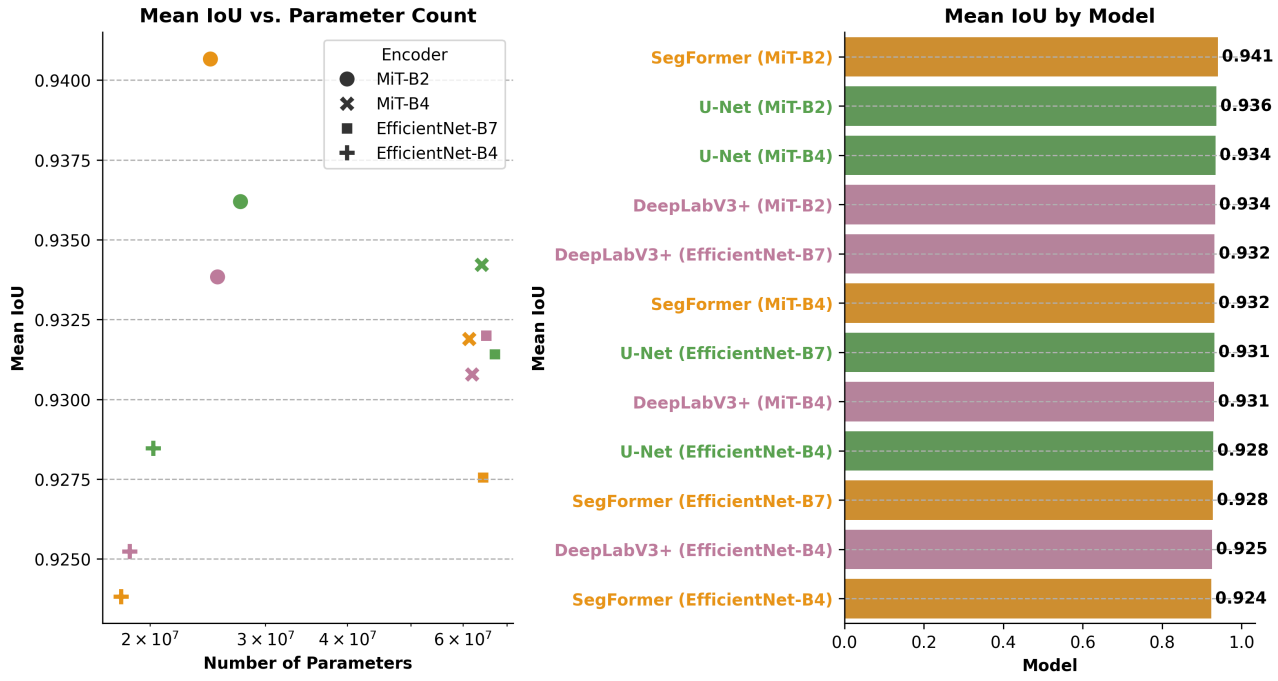


Figure 4.3: Binary classification performance (Water vs. Non-water) and architectural efficiency. The right panel displays the Mean IoU on a standard 0-1 scale, highlighting the generalized predictive saturation across all models. The left panel maps the Mean IoU against the total parameter count, demonstrating that highly efficient encoders (∼25M parameters, such as MiT-B2) systematically outperform their heavier, overparameterized counterparts (>60M parameters).

4.3 Geometric Evaluation: Cross-shore Offsets

4.3.1 Overarching Family Performance (Index-based vs. Deep Learning)

To establish a macroscopic comparison between the two fundamental approaches, the cross-shore extraction errors (offsets) were initially aggregated by model family across both study areas. Figure 4.4 (left panel) presents the overall performance distributions, revealing a decisive superiority of the Deep Learning architectures over the traditional spectral baselines.

Globally, the DL-based models achieved a median offset of 5.56 m, nearly halving the 10.02 m median error recorded by the Index-based methods. Beyond the absolute magnitude of the error, the variance showcases an even more critical disparity. The interquartile range (IQR) for the DL family is remarkably tight (7.42 m), whereas the traditional indices exhibit a widespread error distribution (IQR of 21.08 m). This indicates that, regardless of the specific architecture or thresholding technique applied, semantic segmentation models consistently yield highly stable and precise waterline contours, minimizing severe extraction failures.

However, the overarching advantage of the Deep Learning approach becomes most apparent when the error distributions are disaggregated by study area (Figure 4.4, right panel). This spatial breakdown directly tests the models against the morphological complexities established in Section 4.1. In Section 4.3.3, these differences will be further analyzed in detail.

For the Index-based family, the extraction performance is highly site-dependent. At the relatively sheltered, golden-sanded baseline of El Puntal, the indices perform reasonably well, maintaining a median offset of 7.81 m and a relatively controlled 75th percentile (Q3: 16.69 m). In stark contrast, when deployed over the stress-test environment of Salinas, the traditional algorithms experience a severe degradation. The median error completely doubles to 16.32 m, and the distribution stretches massively, with the Q3 reaching 51.79 m. This severe structural downgrade confirms that pixel-by-pixel spectral formulations are fundamentally vulnerable to the radiometric ambiguity caused by dark carbonaceous sand and extensive wave foam.

Conversely, the Deep Learning models demonstrate an outstanding geographic invariance. At El Puntal, the DL family achieves a median error of 5.47 m. When subjected to the challenging conditions of Salinas, the performance remains virtually unchanged, with a median offset of 5.72 m and a strictly bounded upper quartile (Q3: 10.53 m). This remarkable consistency proves that advanced neural networks successfully leverage broader spatial context and structural patterns. By understanding the semantic relationship between the breaking waves, the wet sand, and the ocean, the DL architectures effectively bypass the spectral limitations that severely penalize traditional thresholding techniques in highly dynamic macrotidal environments.

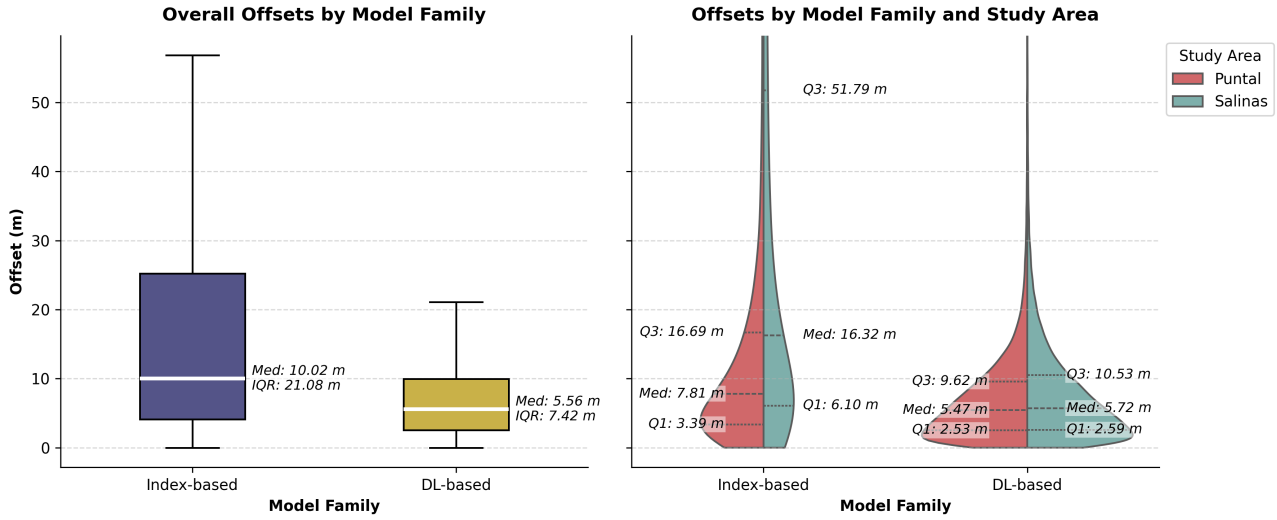


Figure 4.4: Distribution of cross-shore extraction offsets by model family. The left panel (Boxplot) displays the aggregated global performance, highlighting the median and interquartile range (IQR). The right panel (Violin plot) disaggregates the errors by study area (El Puntal vs. Salinas), emphasizing the severe performance degradation of index-based methods in complex environments compared to the geographic robustness of DL models.

4.3.2 Intra-Family Model Assessment and Spatial Bias

Having established the overarching superiority of the Deep Learning approach, it is necessary to intrinsically evaluate the specific configurations within each family. This intra-family assessment identifies which architectural combinations and thresholding techniques yield the highest predictive power, utilizing the Mean Absolute Error (MAE) as the primary evaluation metric.

Regarding the Index-based models, the error matrix reveals an extreme sensitivity to the chosen binarization threshold. The performance of a single spectral index can fluctuate drastically depending on the thresholding algorithm applied. For instance, MNDWI suffers a massive degradation using a Zero threshold (MAE: 48.35 m) but improves significantly with the Local Minimum (LM) Otsu (17.34 m). Conversely, NDWI performs reasonably with a Zero cutoff (20.22 m) but completely fails under LM Otsu (48.43 m). Standard Otsu consistently yields poor results across most indices, struggling with the multimodal histograms of macrotidal beaches. Notably, the optimal configuration within this family is the subtractive SCoWI coupled with the LM Otsu threshold, achieving a MAE of 8.71 m. This result empirically validates the SCoWI's specific formulation designed to suppress wave foam and intertidal noise, provided it is paired with a context-aware thresholding technique.

In stark contrast to the volatility of the traditional indices, the Deep Learning matrix exhibits remarkable algorithmic stability. Regardless of the architecture or encoder chosen, all DL configurations constrained their average error to a tight operational band between 7.27 m and 8.54 m. Within this high-performing group, the U-Net architecture consistently outperformed SegFormer and DeepLabV3+. Most importantly, the results confirm the hybridization hypothesis proposed in the methodological framework (Section 3.5). Combining the global-context awareness of the hierarchical Vision Transformers (MiT-B2 and MiT-B4) into the precise local-feature CNN decoder of U-Net yielded the absolute best global performance. Specifically, the U-Net paired with the MiT-B2 encoder achieved the lowest recorded error of the entire study, with a MAE of 7.27 m. Interestingly, a nuanced pattern emerges in this matrix regarding model capacity and feature extraction depth. With the exception of DeepLabV3+, which marginally benefited from increased parameterization, both the U-Net and SegFormer architectures systematically yielded better geometric accuracy when paired with lighter encoders, such as EfficientNet-B4 and MiT-B2, rather than heavier, more complex counterparts, such as EfficientNet-B7 and MiT-B4. This suggests that excessively deep encoders may induce a degree of spatial overfitting or loss of the high-resolution features necessary for precise shoreline delineation, particularly in the context of the complex coastal environments evaluated in this study.

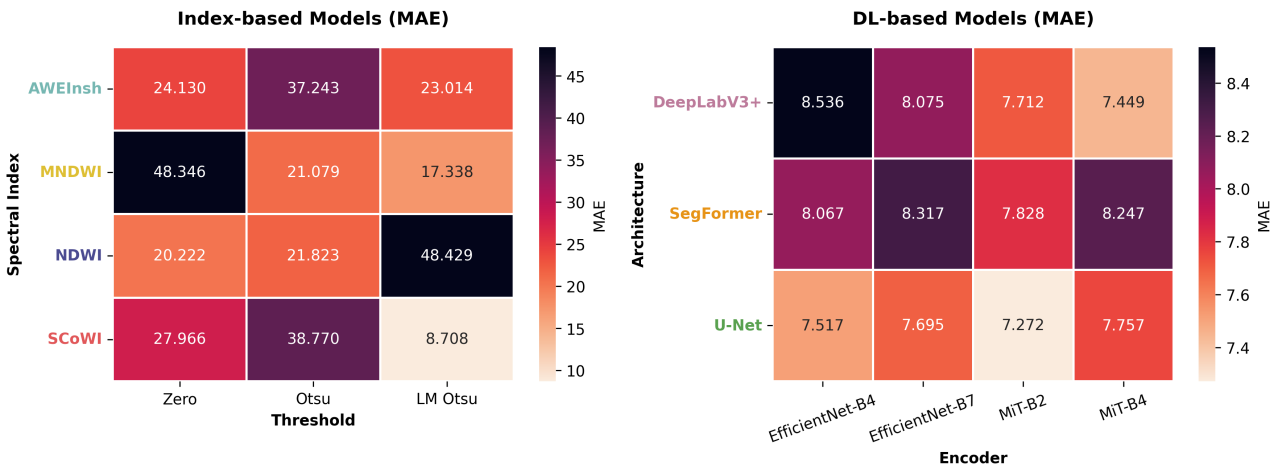


Figure 4.5: Mean Absolute Error (MAE) heatmaps for intra-family configuration assessment. The left panel details the performance of spectral indices across different thresholding techniques, highlighting extreme variance. The right panel displays the Deep Learning configurations, demonstrating high stability and the superiority of the U-Net (MiT-B2) hybrid architecture.

Beyond the absolute magnitude of the extraction error, it is critical for coastal engineering applications to understand the orientation of this displacement. To assess whether the models exhibit a systematic spatial bias (tending to delineate the waterline either too far inland or too far seaward), the cross-shore predictions (SDW) were mapped against the ground-truth reference (RWL) using hexbin density plots.

The hexbin distributions for traditional spectral indices (Figure 4.6) reveal extreme spatial dispersion and distinct systematic failure modes. The Coefficient of Determination (R^2) reflects this volatility, ranging from a low 0.31 (MNDWI Zero) to a highly correlated 0.91 (SCoWI LM Otsu). Visually, the vast majority of spectral combinations exhibit a pronounced *Landward* bias, characterized by large negative Mean Error (Bias) values. For instance, MNDWI (Zero) and AWEInsh (Otsu) yielded biases of -47.87 m and -33.49 m, respectively. This severe landward offset demonstrates that standard indices systematically misclassify the saturated intertidal zone as water, effectively tracing the boundary between dry and wet sand rather than the true instantaneous waterline.

Conversely, NDWI (particularly when coupled with LM Otsu) exhibits an extreme *Seaward* bias, recording a massive positive displacement of +46.18 m. This behavior aligns with the well documented vulnerability of the NDWI to breaking wave foam, causing the algorithm to falsely track the outer edge of the surf zone instead of the shoreline. The only robust exception within this traditional family is SCoWI (LM Otsu). It successfully clusters the bulk of its predictions along the 1:1 agreement bisector, achieving an R^2 of 0.91 and a minimal landward bias of -2.75 m.

In stark contrast, the hexbin density plots for the Deep Learning models (Figure 4.7) display very little spatial dispersion, visually confirming their structural robustness. The visual uniformity across the panels is supported by all evaluated DL configurations achieving a very high $R^2 \geq 0.96$, tightly positioning on the 1:1 bisector line. Unlike the index-based methods, the semantic segmentation architectures completely eliminate the severe landward (wet sand) and seaward (foam) failure modes.

The statistical metrics reveal a highly controlled, very slight *Seaward* bias across all DL architectures, ranging narrowly from +3.20 m (DeepLabV3+ EfficientNet-B4) to +5.48 m (SegFormer MiT-B4). This indicates a minimal, systematic tendency of the DL models to delineate the waterline marginally closer to the swash zone, in contrast to the traditional baseline with its massive errors. Ultimately, the visual and statistical indistinguishability among the DL hexbins reaffirms that these neural networks have successfully learned the underlying semantic context of the coastal boundary, effectively rendering the extraction process immune to the spectral noise of macrotidal environments.

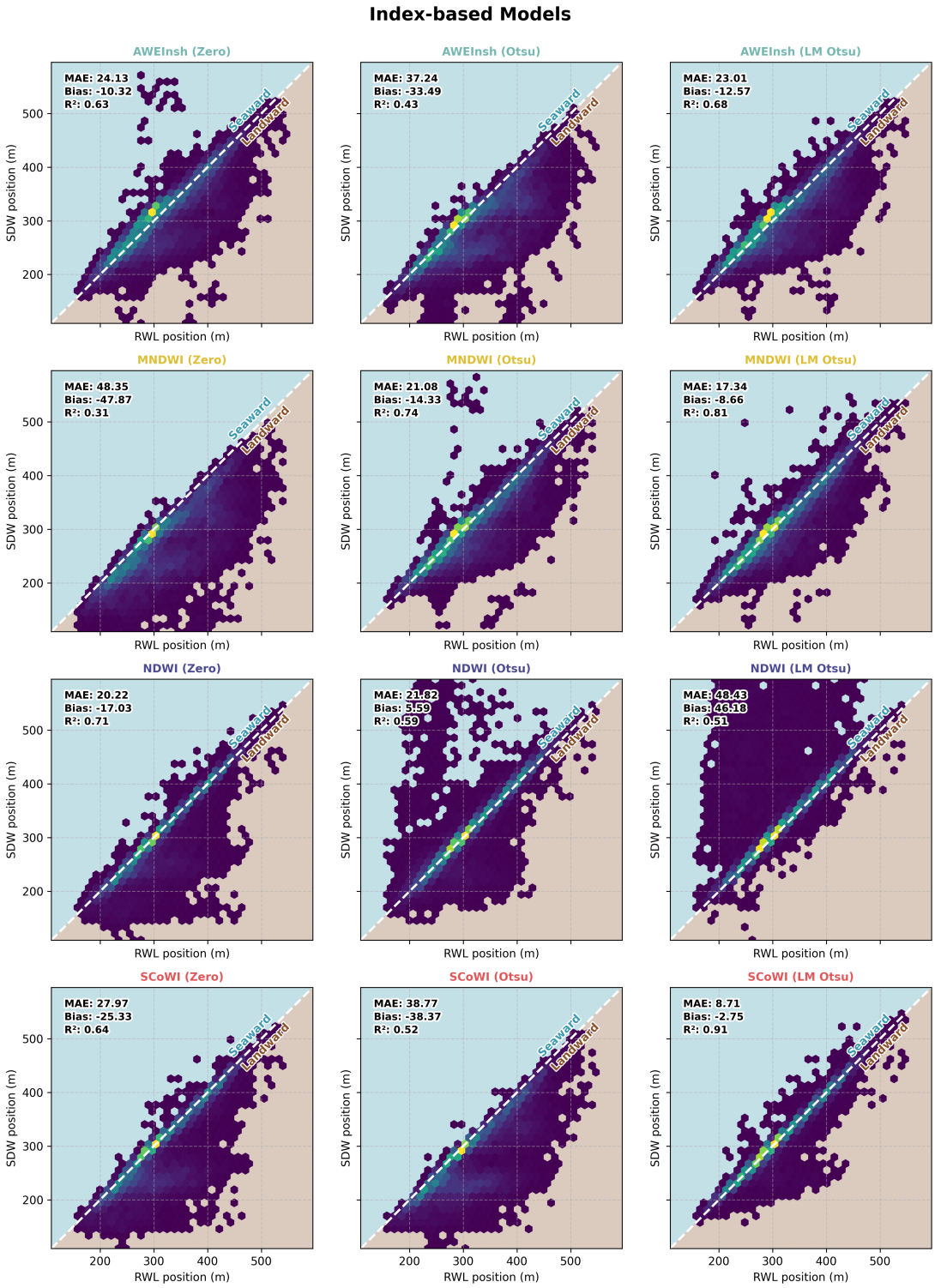


Figure 4.6: Hexbin error distributions for Index-based models. The plots map the predicted shoreline position (SDW) against the reference waterline (RWL). The dashed white line represents the 1:1 perfect agreement. Most indices show a severe Landward bias (wet sand confusion), while NDWI shows a Seaward bias (foam confusion). SCoWI (LM Otsu) stands out as the most accurate traditional method.

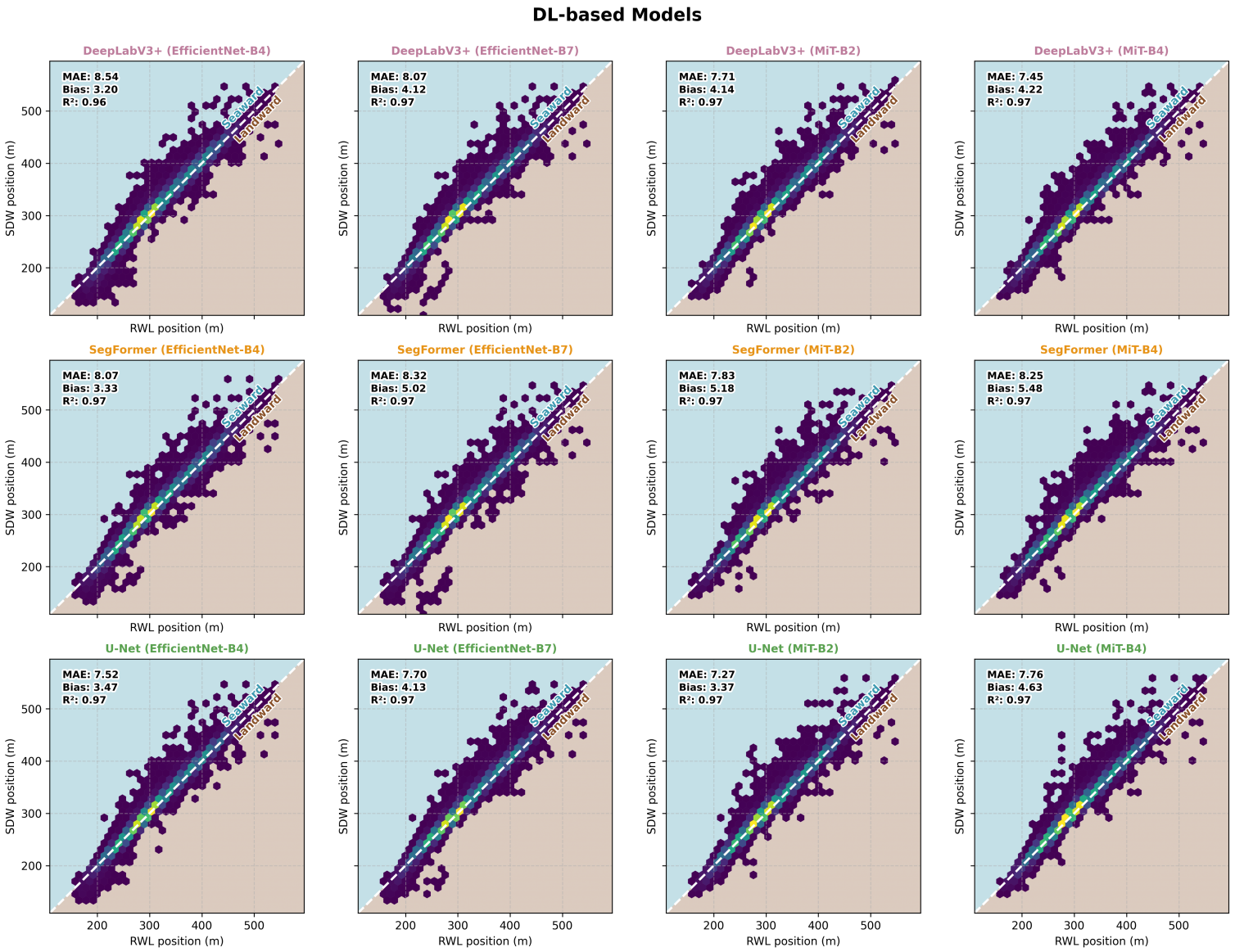


Figure 4.7: Hexbin error distributions for Deep Learning semantic segmentation models. In contrast to the traditional baseline, all DL configurations demonstrate minimal dispersion, uniformly tight clustering around the 1:1 bisector ($R^2 \geq 0.96$), and the complete elimination of extreme cross-shore biases.

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4.3.3 Spatial and Environmental Drivers of Error

Spatial Variability Along the Coastline:

Before evaluating the hydrodynamic forcing, it is essential to examine the spatial variability of the extraction error along the coastline. By analyzing the distribution of cross-shore offsets grouped by `transect_id`, specific morphological, geological, or anthropogenic features that locally disrupt model performance can be identified.

Figure 4.8 details this alongshore variability for the Index-based models. At El Puntal, a distinct spatial pattern emerges: the first 15 transects (from W to E) exhibit error metrics (median, third quartile, and maximum bounds) notably below the overall site average. This localized increase in accuracy is driven by two main factors. First, the initial ≈ 8 transects correspond to the navigation channel of the Santander Bay, an area subjected to periodic dredging operations. Second, this entire initial stretch is highly sheltered from incident wave energy, minimizing foam-induced spectral noise. From transect 15 onwards, the offsets closely align with the overall median, until the final 5 transects, where a sudden spike in error is recorded. This localized degradation coincides with a slight shift in the coastline orientation and the proximity to a small stream mouth, which likely introduces spectral confusion due to localized moisture and water discharge.

At Salinas, the traditional spectral indices also display significant alongshore volatility. The initial transects record lower-than-average offsets; this physically corresponds to the narrowest sector of the beach where high tides frequently inundate the entire dry sand area, paradoxically restricting the physical space for the models to commit severe landward misclassifications. Conversely, a distinct cluster of elevated errors emerges between transects 12 and 21. This specific coastal stretch is backed by high-rise urban infrastructure. During the Sentinel-2 mid-morning overpass, these buildings project prominent shadows across the intertidal zone, severely confusing the pixel-by-pixel radiometric thresholding of traditional indices. Finally, the last two transects, located immediately adjacent to the eastern inlet jetty, show reduced offsets, likely due to the constrained morphodynamic mobility of the sand volume trapped against this rigid coastal structure.

In sharp contrast to the baseline, the spatial variability across transects for the Deep Learning models (Figure 4.9) is remarkably lower, visually corroborating their geographic robustness. At El Puntal, there is a marginal reduction in offsets between transects 3 and 10, but the magnitude of this variation is virtually negligible compared to the volatility of the index-based models. Similarly, at Salinas, the DL predictions remain stable across the entire beach longitude, effectively bypassing the severe urban shadow artifacts seen in spectral indices. The only notable, albeit minor, anomaly occurs in the last four transects near the Salinas jetty. Here, the DL models register slightly higher offsets relative to their internal overall median.

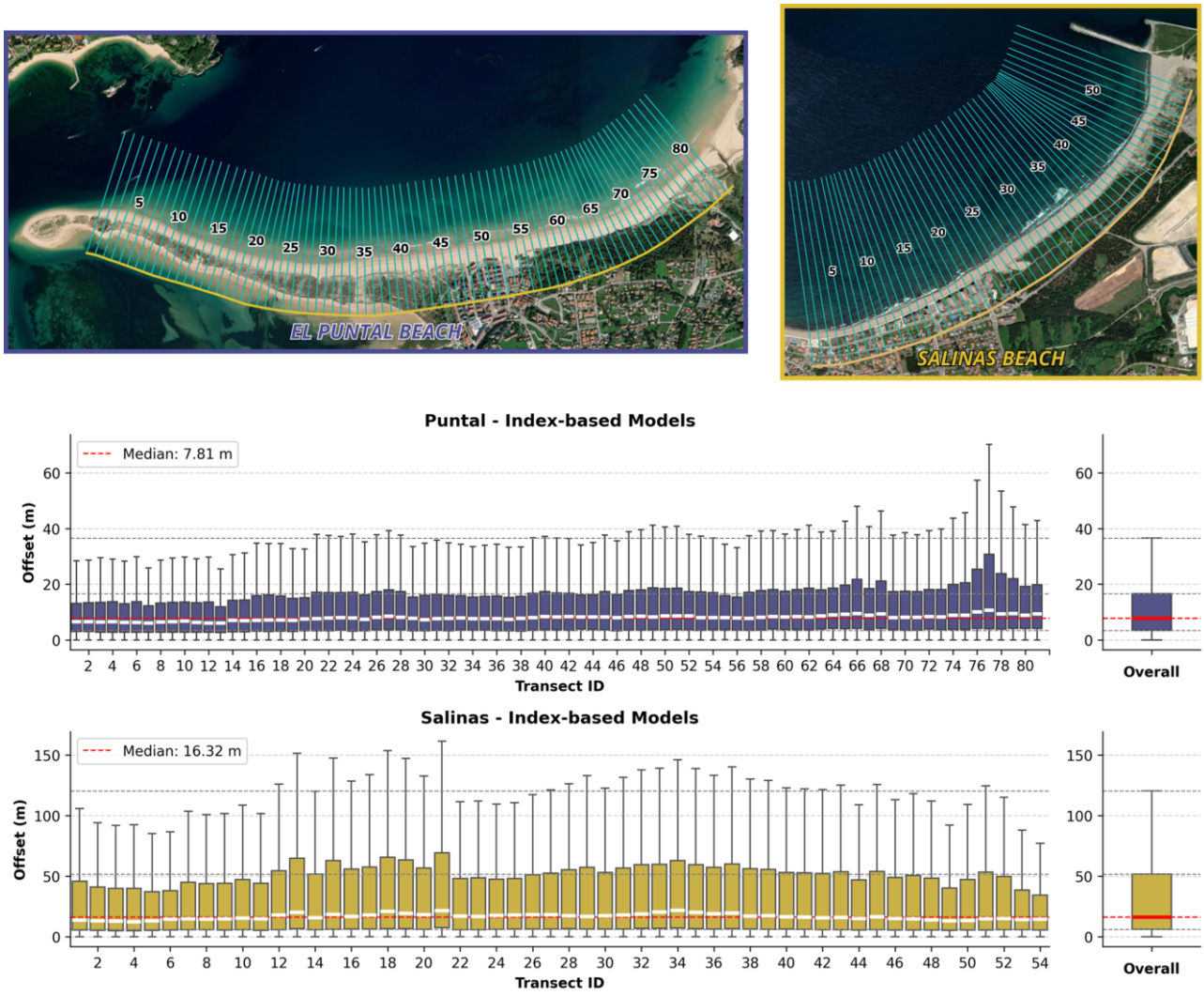


Figure 4.8: Alongshore variability of extraction offsets for Index-based models. The upper map panels reference the geographical location of the cross-shore transects. The boxplots show the error distribution per `transect_id` compared to the overall site performance (right sub-panels).

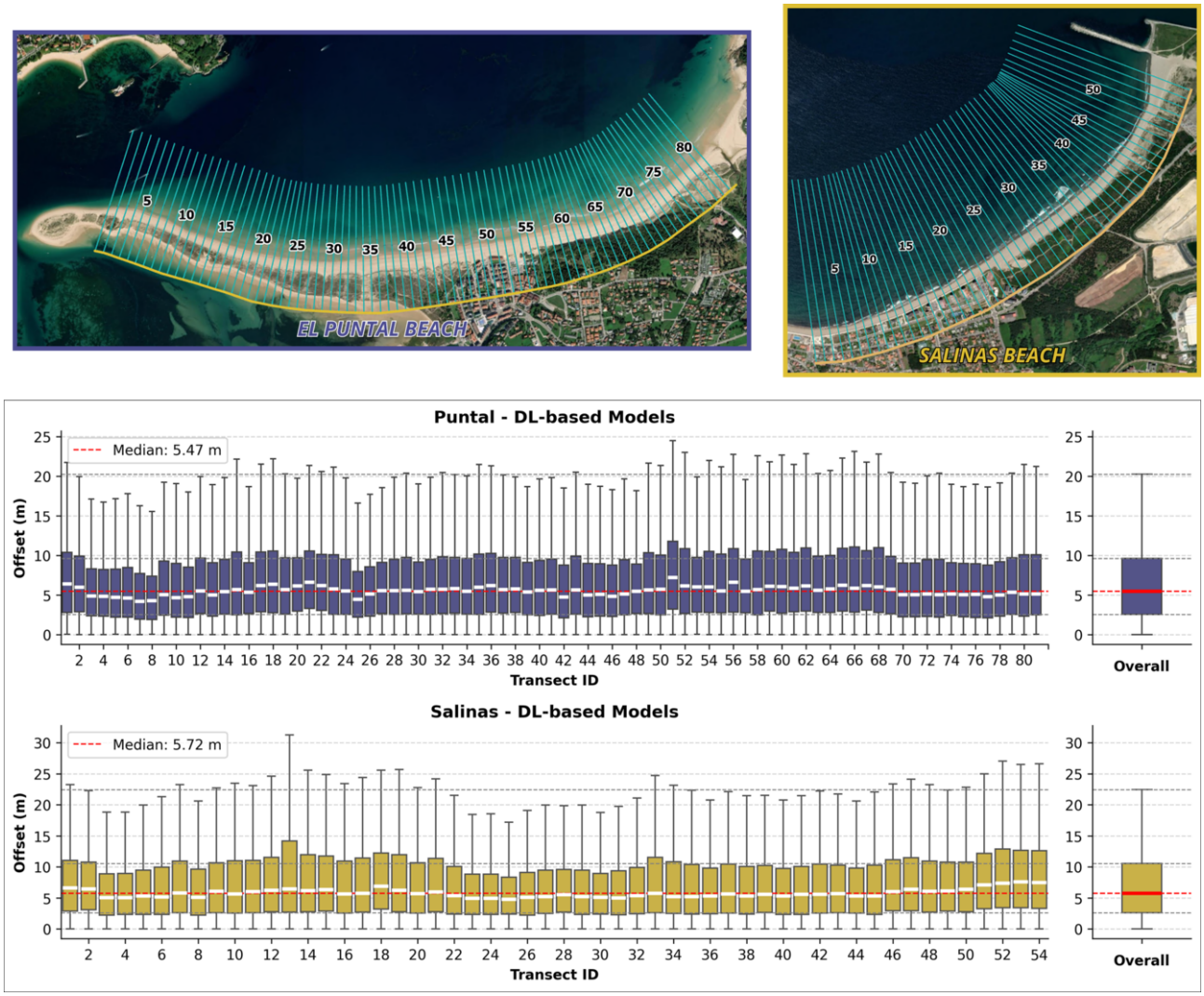


Figure 4.9: Alongshore variability of extraction offsets for Deep Learning models. The upper map panels reference the geographical location of the cross-shore transects. The boxplots show the error distribution per `transect_id` compared to the overall site performance (right sub-panels).

Hydrodynamic Forcing, Tide and Wave Energy:

Beyond spatial artifacts, the temporal variability forced by hydrodynamic conditions, specifically astronomical tide and significant wave height (H_s) was evaluated. To isolate the environmental sensitivity of the models from their baseline errors, the performance was analyzed using the relative percentage change ($\Delta\%$) of the offset within each hydrodynamic quartile against the overall median of each specific study site.

The tide quartile analysis reveals a critical, structurally consistent phenomenon: the lowest performance across all models occurs during low tide (Q1), rather than high tide (Q4). For the Index-based models, this degradation is catastrophic in the Salinas stress-test environment, where the Q1 relative offset spikes by +312.9% (reaching an absolute median error of 67.38

m). Physically, low tide in macrotidal beaches exposes a vast intertidal zone characterized by water-saturated sand, complex rhythmic forms, and remnant pooling. This flat, dark, and highly reflective wetted surface critically confuses pixel-based thresholding algorithms. Conversely, during high tide (Q4), the complex intertidal zone is entirely submerged, leaving a starker dry sand-water boundary that improves the index-based relative performance by -51.6% in Salinas and -22.0% in El Puntal.

The Deep Learning models exhibit a conceptually similar sensitivity profile to the tidal cycle, confirming that the exposure of the intertidal zone is an inherently challenging feature even for human annotators. However, the magnitude of their sensitivity is exceptionally controlled. In Salinas, the DL Q1 relative downgrade is only +30.2%, which translates to a reasonably operational absolute median offset of just 7.44 m. This demonstrates that while the neural networks detect the physical ambiguity of the wet sand, their spatial context awareness prevents the catastrophic delineations that plague traditional indices.

Regarding wave energy forcing, a direct correlation between significant wave height (H_s) and extraction error is observed. For the Index-based models, transitioning from calm seas (Q1) to storm conditions (Q4) steadily increases the relative error in both El Puntal (from -7.0% to +16.9%) and Salinas (from -16.5% to +25.0%). This deterioration is a direct consequence of extensive white water foam and wave run-up physically widening the swash zone, which the spectral indices misclassify.

The DL-based architectures once again demonstrate superior resilience against wave forcing. At El Puntal, the models remain completely stable regardless of the wave height, fluctuating minimally. In Salinas, the presence of extreme wave conditions (Q4) induces a slight relative increase of +18.0%. Nevertheless, this relative bump corresponds to an absolute median error of only 6.74 m, confirming that the semantic segmentation networks can accurately track the underlying shoreline position regardless of extreme surf zone hydrodynamics.

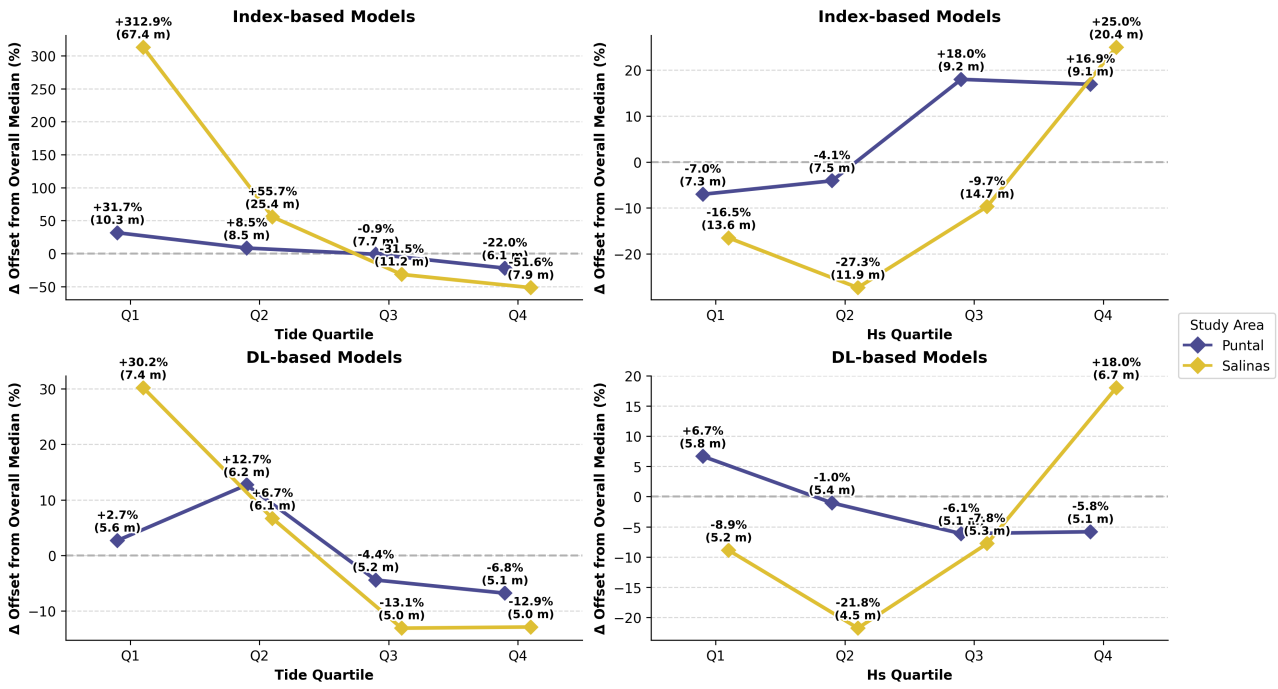


Figure 4.10: Relative performance sensitivity to hydrodynamic forcing. The plots display the percentage change in the median offset (Δ%) relative to the overall site median across astronomical tide (left) and significant wave height (right) quartiles. Absolute median values are annotated for each quartile to contextualize the operational impact of these relative changes.

4.3.4 Cross-Site Generalization and Robustness

A critical requirement for any automated waterline extraction pipeline is its ability to operate reliably across varying geographical environments without requiring site-specific recalibration. To evaluate this spatial robustness, the cross-site generalization was quantified by calculating the difference in Mean Absolute Error (Δ MAE) between the baseline environment (El Puntal) and the morphodynamic stress-test site (Salinas) for each model. A Δ MAE close to zero indicates perfect generalization, whereas large negative values denote severe performance degradation when transitioning to the complex beach.

As visualized in the lollipop charts (Figure 4.11), the traditional Index-based family struggles significantly with cross-site transferability, recording a massive mean degradation of 23.16 m. This confirms that spectral thresholding is highly sensitive to local radiometric conditions. NDWI (LM Otsu) suffers a catastrophic collapse, deteriorating by 79.04 m in Salinas, further proving its severe vulnerability to the wave foam prevalent in higher-energy macrotidal shores. Even the more robust thresholding configurations, such as MNDWI (Zero) and NDWI (Zero), degrade by 19.30 m and 30.80 m, respectively. The only acceptable outlier within this family is, once again, the SCoWI coupled with LM Otsu, which manages to constrain its degradation to 5.89 m. While commendable for a traditional index, this value still represents a substantial loss of precision.

In striking contrast, the Deep Learning family demonstrates an extraordinary level of cross-site robustness. The models exhibit a near-perfect generalization, with a mean Δ MAE of merely -0.68 m across all tested architectures. This structural stability implies that the neural networks degrade by less than a tenth of the native Sentinel-2 pixel size (10 m) when facing the dark sand, shadows, and complex intertidal morphology of Salinas.

Within this highly robust group, several configurations achieved virtually identical performance across both beaches. SegFormer (EfficientNet-B7) and DeepLabV3+ (MiT-B2) recorded Δ MAEs of just -0.03 m and -0.09 m, respectively. Even the U-Net (MiT-B2), which was previously identified as the top performer in terms of absolute MAE (Section 4.3.2), generalized exceptionally well with a minor penalty of -0.96 m. Even the "worst-generalizing" DL model, DeepLabV3+ (EfficientNet-B4), only degraded by 1.57 m.

These results conclusively prove that semantic segmentation models do not overfit to the local radiometry of golden, semi-sheltered beaches. Instead, they successfully abstract the universal semantic features of the land-water boundary, rendering them highly reliable for large-scale, automated coastal monitoring in heterogeneous environments.

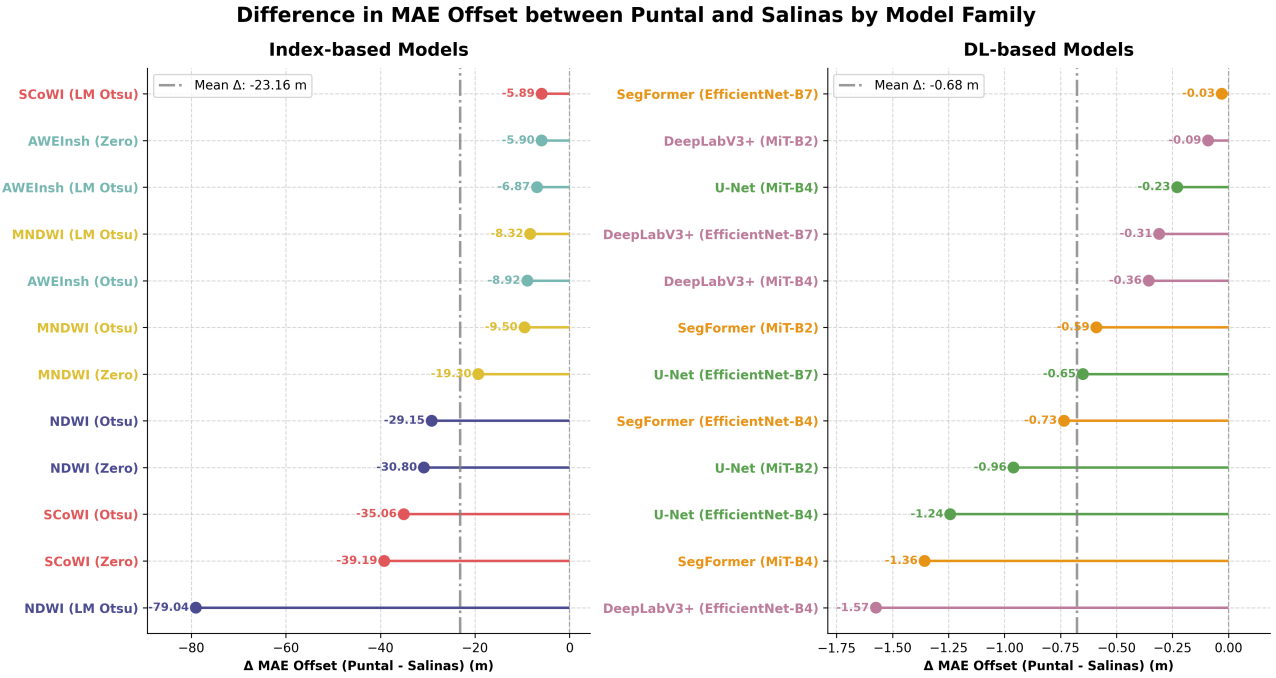


Figure 4.11: Cross-site generalization evaluated through the difference in Mean Absolute Error (Δ MAE) between El Puntal and Salinas. Values closer to the zero vertical line indicate higher robustness to changing environmental conditions. The DL-based models (right panel) demonstrate near-perfect generalization compared to the highly site-dependent Index-based baseline (left panel).

4.3.5 Final Multi-Criteria Ranking

While the preceding subsections isolated specific model vulnerabilities, such as structural bias, spatial artifacts, and hydrodynamic forcing, a holistic evaluation requires synthesizing all geometric error metrics into a unified consensus. To definitively determine the optimal architecture for automated supratidal shoreline extraction, a multi-criteria ranking was established based on the aggregated performance across six robust statistical indicators: MAD, IQR, MAE, Bias, RMSE, and R^2.

The overall metrics heatmap (Figure 4.12) illustrates the absolute performance values for all 24 evaluated configurations. The dichotomy between the two model families is numerically remarkable, with the Deep Learning architectures monopolizing the lowest error bands across almost all indicators. However, a highly notable anomaly exists within the traditional baseline: SCoWI (LM Otsu). This specific index achieved the absolute best global Median Absolute Deviation (MAD: 4.90 m) and the most neutral mean error (Bias: -2.75 m), outperforming even the neural networks in these specific central-tendency metrics.

Despite this impressive baseline precision, SCoWI (LM Otsu) suffers a significant penalty in its RMSE (16.67 m) and MAE (8.71 m) compared to the DL models. Physically, this indicates that while the SCoWI formulation successfully delineates the waterline accurately under optimal conditions, it remains vulnerable to severe, localized extraction failures that heavily penalize the quadratic and absolute mean errors. Conversely, the Deep Learning models exhibit an extraordinary balance, maintaining tight IQRs (all ≤ 11.95 m) and extremely low RMSE values, proving their immunity to catastrophic extraction outliers.

To distill these absolute values into a definitive hierarchy, a bump chart (Figure 4.13) was generated by computing the rank of each model per metric and calculating their arithmetic mean. This comprehensive ranking system successfully identifies the top three performing architectures, crowning them as the definitive state-of-the-art for macrotidal environments:

- ⋆ **1st Place: U-Net (MiT-B2) [Average Rank: 2.33].** This hybrid configuration is the undisputed global champion. By combining the precise spatial localization of the U-Net CNN decoder with the global context awareness of the MiT-B2 Vision Transformer, it secured the absolute #1 rank in both MAE (7.27 m) and RMSE (11.35 m). This architecture provides the best overall balance between accuracy and outlier suppression.
- ⋆⋆ **2nd Place: DeepLabV3+ (MiT-B4) [Average Rank: 3.17].** This highly complex model proved to be the most consistent architecture of the study. It achieved the absolute #1 rank in IQR (9.50 m) and R^2 (0.97), and the #2 rank in MAE (7.45 m). Its performance demonstrates that advanced atrous spatial pyramid pooling (ASPP) paired with a deep transformer encoder yields incredibly stable waterline contours.
- ⋆⋆⋆ **3rd Place: U-Net (EfficientNet-B4) [Average Rank: 3.33].** Proving that pure Convolutional Neural Networks are still highly competitive, this lightweight U-Net secured the third position globally. It ranked #2 in R^2 and #3 in both MAD and RMSE, offering an outstanding trade-off between architectural efficiency and geometric precision.

Ultimately, this multi-criteria synthesis conclusively answers the core research question: Deep Learning semantic segmentation models (specifically hybrid U-Net architectures) outperform traditional spectral indices, providing a structurally robust, environmentally resilient, and geographically invariant solution for satellite-derived waterline extraction.

Overall - Model Metrics

	MAD	IQR	MAE	Bias	RMSE	R^2
AWEInsh (Zero)	15.12	33.90	24.13	-10.32	38.40	0.63
AWEInsh (LM Otsu)	13.88	34.48	23.01	-12.57	36.00	0.68
AWEInsh (Otsu)	19.90	54.36	37.24	-33.49	57.03	0.43
MNDWI (Zero)	31.06	53.18	48.35	-47.87	71.27	0.31
MNDWI (LM Otsu)	10.65	23.57	17.34	-8.66	27.08	0.81
MNDWI (Otsu)	11.76	29.56	21.08	-14.33	33.99	0.74
NDWI (Zero)	10.07	25.13	20.22	-17.03	32.36	0.71
NDWI (LM Otsu)	7.28	74.90	48.43	46.18	93.06	0.51
NDWI (Otsu)	8.36	15.82	21.82	5.59	45.49	0.59
SCoWI (Zero)	18.07	38.89	27.97	-25.33	40.63	0.64
SCoWI (LM Otsu)	**4.90**	9.96	8.71	**-2.75**	16.67	0.91
SCoWI (Otsu)	25.31	47.34	38.77	-38.37	55.62	0.52
DeepLabV3+ (EfficientNet-B4)	6.13	11.95	8.54	3.20	12.62	0.96
DeepLabV3+ (EfficientNet-B7)	5.70	10.66	8.07	4.12	12.33	0.97
DeepLabV3+ (MiT-B2)	5.53	10.16	7.71	4.14	11.81	0.97
DeepLabV3+ (MiT-B4)	5.36	**9.50**	7.45	4.22	11.49	**0.97**
SegFormer (EfficientNet-B4)	5.88	10.96	8.07	3.33	11.96	0.97
SegFormer (EfficientNet-B7)	6.08	10.32	8.32	5.02	12.47	0.97
SegFormer (MiT-B2)	5.44	9.77	7.83	5.18	12.13	0.97
SegFormer (MiT-B4)	6.01	10.31	8.25	5.48	12.32	0.97
U-Net (EfficientNet-B4)	5.26	9.84	7.52	3.47	11.53	0.97
U-Net (EfficientNet-B7)	5.46	10.14	7.70	4.13	11.75	0.97
U-Net (MiT-B2)	5.11	9.81	**7.27**	3.37	**11.35**	0.97
U-Net (MiT-B4)	5.55	9.85	7.76	4.63	12.27	0.97

Figure 4.12: Overall multi-criteria performance heatmap. The matrix displays the absolute values for MAD, IQR, MAE, Bias, RMSE, and R^2 across all evaluated models. The best absolute value for each metric is highlighted in bold, showcasing the Deep Learning dominance and the anomalous central-tendency precision of the SCoWI (LM Otsu) index.

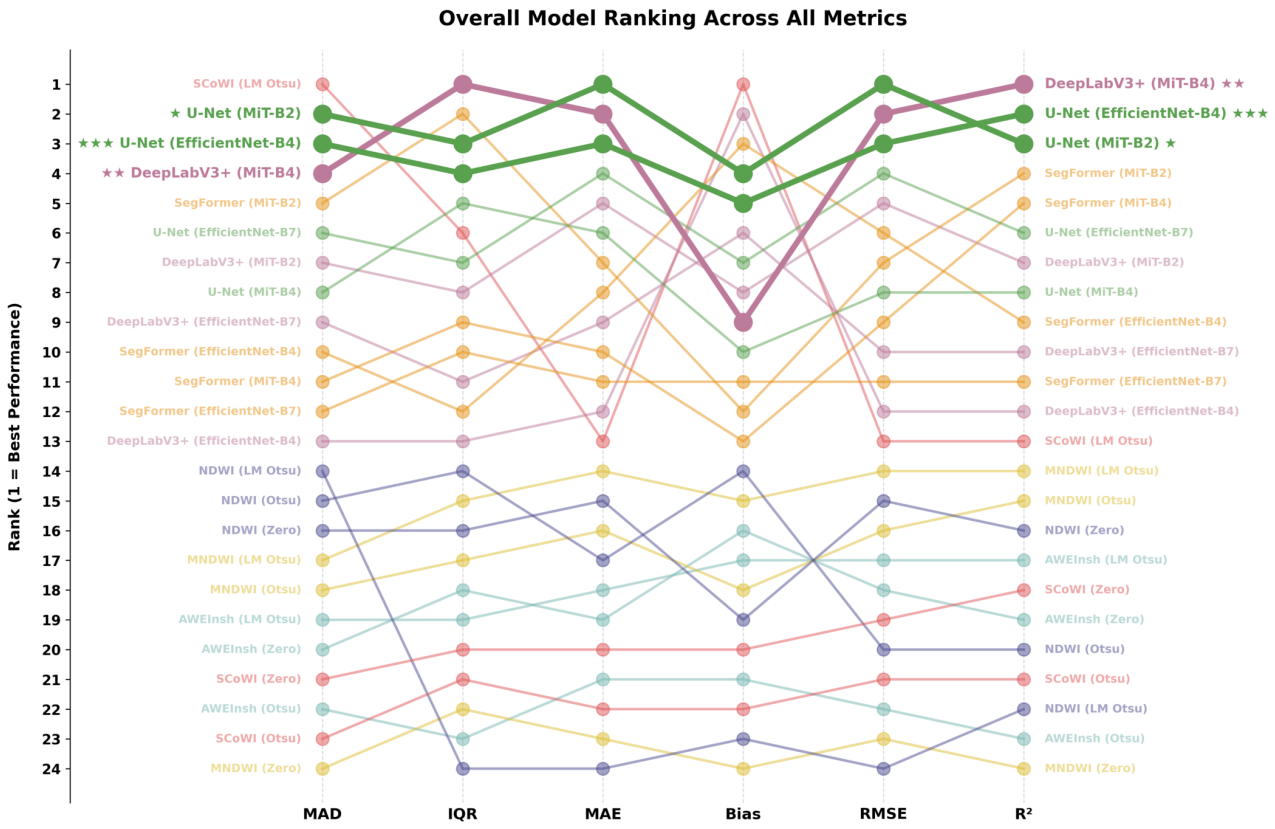


Figure 4.13: Comprehensive bump chart ranking based on the six geometric and statistical metrics (1 = Best Performance). The arithmetic mean of these ranks dictates the global hierarchy, highlighting the Top 3 global architectures (U-Net MiT-B2, DeepLabV3+ MiT-B4, and U-Net EfficientNet-B4).

Chapter 5

Conclusions and Future Works

5.1 Conclusions

5.1.1 Synthesis of Results and Key Findings

This study set out to rigorously evaluate and benchmark the performance of traditional spectral water indices against state-of-the-art Deep Learning (DL) semantic segmentation models for the task of satellite-derived waterline extraction in highly dynamic macrotidal environments.

The results unequivocally demonstrate that Deep Learning architectures provide a structurally robust, environmentally resilient, and geographically invariant solution, drastically outperforming the traditional index-based baseline. While traditional indices (such as NDWI, MNDWI, and AWEInsh) showed acceptable performance in sheltered and golden-colored sand beach (El Puntal), they suffered catastrophic degradation when subjected to the stress-test of Salinas beach. The primary driver of this failure was the spectral ambiguity introduced by the vast expanses of dark, saturated sand exposed during low tide, which pixel-based thresholding algorithms systematically misclassified as water, leading to massive landward biases (e.g., median absolute offsets exceeding 67 m in extreme cases). Furthermore, traditional methods proved highly vulnerable to local radiometric noise, such as urban shadows and breaking wave foam.

Conversely, the DL models successfully abstracted the general semantic features of the land-water boundary. By leveraging spatial context, architectures like U-Net, DeepLabV3+, and SegFormer effectively differentiated between shallow water, wave foam, and wet sand. Consequently, they maintained a remarkable cross-site generalization, degrading by less than a tenth of the native pixel size (Δ MAE of -0.68 m) when transitioning to the more complex environment.

A valuable finding of this research was the inverse relationship observed between network parameterization and geometric precision. It was initially expected that heavier, more complex encoders (such as EfficientNet-B7 or MiT-B4) would yield superior performance. However, the results revealed that lighter encoders (specifically MiT-B2 and EfficientNet-B4) consistently achieved higher sub-pixel accuracy and tighter extraction contours. This demonstrates that for

the highly specific task of waterline delineation, overparameterization induces spatial overfitting, sacrificing the high-resolution localization required to trace a sharp boundary. Alongside this efficiency, the study highlighted the effectiveness of architectural hybridization. The results demonstrated that pairing the Vision Transformer encoders with the Convolutional Neural Network decoders creates a highly synergistic framework. Ultimately, this hybrid approach (specifically the U-Net architecture paired with the MiT-B2 encoder) emerged as the definitive state-of-the-art, achieving the lowest overall errors (MAE: 7.27 m) and the highest resistance to extraction outliers.

5.1.2 Achievement of Objectives

The primary objective of this Master's Thesis, that is, to establish a comprehensive, data-driven benchmarking framework comparing traditional and AI-based methods for continuous shoreline monitoring in complex coastal zones has been fully achieved. Furthermore, the specific secondary objectives were successfully met: a high-quality, multi-class Ground Truth dataset was generated; model degradation under adverse hydrodynamic conditions (such as low-tide exposure and high wave energy) was rigorously quantified; the cross-site transferability between baseline and stress-test coastal environments was validated; and the computational efficiency trade-offs among the different extraction architectures were systematically evaluated.

5.1.3 Project Planning and Methodological Evolution

The overall execution of the project successfully adhered to the initially established five-month schedule. The core planned methodology, encompassing satellite data acquisition, cross-shore spatial evaluation, and hydrodynamic sensitivity analysis, proved to be highly adequate for the intended research goals.

5.1.4 Evaluation of Foreseen Impacts

In accordance with institutional guidelines regarding cross-disciplinary competencies, the execution of this work maintained a neutral to positive footprint across all evaluated dimensions.

Regarding sustainability, the methodology relies on free, open-access satellite Earth Observation data, which effectively negates the need for physical, carbon-intensive topographical field campaigns. While the training of the neural networks incurred a localized energy cost via GPU processing, this was computationally optimized and strictly bounded, resulting in an overall positive environmental balance.

In terms of ethical behavior, social responsibility, and diversity, the project utilized exclusively anonymized, public remote sensing data. No unforeseen negative impacts materialized during the development of this strictly technical framework. The code and models were developed avoiding any implicit biases, ensuring full compliance with the initial assessments and maintaining a neutral societal footprint.

5.2 Future Works

While this study establishes a robust and operational framework for DL-based coastal monitoring, several tasks for future research remain open:

- **Global Expansion of the Training Dataset:** The current models were successfully trained and validated on highly dynamic macrotidal beaches in Northern Spain. Future iterations should incorporate a globally distributed dataset encompassing diverse coastal morphologies to guarantee universal geographic invariance.
- **Sub-pixel Resolution Enhancement:** Exploring advanced super-resolution neural networks over the Sentinel-2 data could further refine the spatial precision of shoreline extraction, potentially enhancing sub-meter accuracy necessary for critical coastal engineering applications.
- **Development of an Open-Source Operational Tool:** Translating the optimal architecture (U-Net MiT-B2) into an accessible, end-to-end open-source software package or cloud-based API, enabling coastal managers and engineers to deploy this technology seamlessly without requiring deep learning expertise.

Chapter 6

Glossary

ASPP Atrous Spatial Pyramid Pooling

AWEInsh Automated Water Extraction Index – No Shadows

CNN Convolutional Neural Network

DL Deep Learning

EO Earth Observation

ESA European Space Agency

GNSS Global Navigation Satellite Systems

GT Ground Truth

HAT Highest Astronomical Tide

H_s Significant Wave Height

HWL High-Water Line

IoU Intersection over Union

IQR Interquartile Range

LAT Lowest Astronomical Tide

LiDAR Light Detection and Ranging

MAD Median Absolute Deviation

MAE Mean Absolute Error

MHW Mean High Water

MiT Mix Transformer

ML Machine Learning

MLP Multilayer Perceptron

MNDWI Modified Normalized Difference Water Index

MSL Mean Sea Level

NDWI Normalized Difference Water Index

NIR Near-Infrared

RMSE Root Mean Square Error

ROI Region of Interest

RTK Real-Time Kinematic

RWL Reference Waterline

SAM Segment Anything Model

SCoWI Subtractive Coastal Water Index

SDW Satellite-Derived Waterline

SWIR Shortwave-Infrared

T_p Peak Wave Period

ViT Vision Transformer

WL Waterline

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